



Survey Paper

Enhancing road safety with machine learning: Current advances and future directions in accident prediction using non-visual data

Albe Bing Zhe Chai^{a,*}, Bee Theng Lau^a, Mark Kit Tsun Tee^a, Chris McCarthy^b^a Faculty of Engineering, Computing and Science, Swinburne University of Technology, Kuching, Sarawak, 93350, Malaysia^b Department of Computer Science and Software Engineering, Swinburne University of Technology, Hawthorn, VIC, 3122, Australia

ARTICLE INFO

Keywords:

Road safety
Road traffic accident
Predictive model
Machine learning
Deep learning

ABSTRACT

Road traffic accident (RTA) poses a significant road safety issue due to the increased fatalities worldwide. To address it, various artificial intelligence solutions are developed to analyze RTA characteristics and make predictions. This study contributed to a systematic review of machine learning (ML) RTA prediction, specifically focusing on the non-visual approaches. It provides insights into recent advancements and introduces potential future research directions for the non-visual approach to better improve road safety. There are 95 studies shortlisted for non-visual ML-based RTA prediction, covering different modeling categories such as risk level, occurrence, frequency, and severity level. The findings revealed that government departments and open-access data portals are the trending sources for RTA datasets, while sources such as sensors and surveys can be used to collect additional RTA information. Moreover, conventional ML approaches are more prominent than statistical regression for RTA prediction due to their ability to handle RTA datasets with complex relationships. Meanwhile, deep learning and hybrid approaches are emerging techniques that are worth investigating in future research. Specifically, hybrid models can boost performance by combining multiple algorithms to focus on different processes such as feature extraction and model tuning. Eventually, several potential directions are highlighted for future research, such as multitask RTA prediction, data quality issues, ethical concerns for real-world implementation, model transferability, and the adoption of a standardized framework for model development. These recommendations are believed to be the catalyst that promotes further research advancement in the context of RTA prediction with non-visual data.

1. Introduction

A road traffic accident (RTA) is defined as a collision involving at least one vehicle with roadside objects or other vehicles and can result in property damage, injuries, or fatalities (Mamo et al., 2023). It is currently a global challenge that causes approximately 1.3 million fatalities annually (WHO, 2022). This is especially significant for children and young adults aged between 5 and 29 years, where RTA is the leading cause of fatalities (WHO, 2018). In addition, the progress in reducing the impact of RTA is different among high-, middle- and low-income countries. There is a limited reduction of RTA fatality rate in most low- and middle-income countries (LMICs) as compared to the high-income countries. Moreover, vulnerable users such as pedestrians, cyclists, two-wheelers, and three-wheelers are the prominent groups who suffer from fatalities due to the imbalanced efforts to provide more protection for them (WHO, 2018). Undoubtedly, the upward trend of

RTA fatalities is a critical challenge that demands dedicated efforts worldwide to formulate appropriate action plans to mitigate the situation.

The United Nations (UN) has attempted to raise road safety awareness globally through Road Safety Weeks, in which several targets and indicators are set to improve road safety levels by 2030 (Hyder et al., 2022). Some of them include the standard technical requirements for constructing new roads, using helmets and seat belts, and reducing RTA due to drunk driving and mobile phone distraction. In addition, another significant target is to reduce the time taken for the emergency response team to recover the traffic at the RTA scene. These targets require substantial research efforts to develop innovative solutions that help improve road safety in different countries. Over the years, researchers have investigated the characteristics of crashes and developed various crash modeling solutions. Furthermore, the advancement in information technology has shifted the research focus from traditional statistics to

* Corresponding author.

E-mail addresses: 104136569@students.swinburne.edu.my, abzchai@swinburne.edu.my (A.B.Z. Chai).<https://doi.org/10.1016/j.engappai.2024.109086>

Received 30 October 2023; Received in revised form 17 June 2024; Accepted 31 July 2024

Available online 11 August 2024

0952-1976/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

artificial intelligence (AI) based solutions. This is driven by the remarkable capabilities of machine learning (ML) approaches to produce promising results that are comparable to traditional methods (Silva et al., 2020).

Road safety research has been found to achieve positive outcomes in high-income countries such as the United States (US), Canada, and the United Kingdom (UK), where the RTA fatality rates in these countries have decreased over the years. The researchers developed various real-time crash prediction models that work with advanced technology such as traffic surveillance cameras, sensory devices, satellites, and graphical information systems (GIS). These advanced systems allowed the development of visual-based solutions to enhance road safety. For instance, visual data such as real-time traffic videos can be used to analyze spatiotemporal features and characteristics. This is useful for object detection and trajectory analysis to identify RTA scenes in real time. However, these advanced solutions are deemed ineffective for most LMICs which are still in the developing phase. These countries might have limited budgets allocated to install a complete and reliable traffic management system. For instance, there is no reliable RTA detection and reporting system in Tanzania (Mrema and Dida, 2020), while the GIS in India is said to be unreliable in generating accurate location data (Bhardwaj and Goswami, 2019). These issues indicate that ML-based solutions with non-visual data could be more feasible for LMICs when non-visual inputs are obtainable in real-time. A reason behind this is that non-visual data (text-based) would typically have a simpler structure than visual data (image-based) and thus, its real-time application would consume less computational power and require lower bandwidth to upload or transfer the data. The higher bandwidth requirement for a visual-based system is also highlighted by R. Liu et al. (2021), where the large-scale implementation can result in high pressure on the bandwidth.

The non-visual solution is a potential direction that is worth investigating. It can be better than the visual approach for some specific tasks. For example, the non-visual solution is superior for analyzing the correlation between the non-visual RTA attributes such as driver's profile, vehicle type, weather conditions, lighting conditions and road geometric conditions. This could be useful for assessing the RTA risk, crash type and severity so that appropriate action plans and strategies can be formulated to assist in the different phases of the disaster management cycle (DMC), covering the preparedness (or prevention), response, recovery, and mitigation (Tay et al., 2022).

In line with the potential of non-visual ML-based road safety solutions for LMICs, it is crucial to review the state-of-the-art research progress so that the prevailing trends and existing gaps are determined to improve the solutions in terms of resources, effectiveness, and accuracy. Moreover, it is beneficial to identify the potential future directions towards a real-time feasible solution with non-visual data for spontaneous RTA analysis and formulation of prevention strategies. To that end, this work is motivated to perform a systematic review under the scope of ML-based RTA prediction models that involve non-visual data.

This work is organized with the following flow. Section 2 discusses some existing reviews related to transportation and road safety modeling. The potential improvements are highlighted and formed the contributions of this work. Then, Section 3 presents the methodology adopted in this work based on Kitchenham's guidelines for systematic review. All findings are analyzed and discussed in Section 4. The findings and discussion are eventually summarized through the conclusion in Section 5.

2. Related work

The deteriorating impact caused by RTA is to be taken seriously to mitigate the increasing trend of fatality rates, especially in LMICs. Researchers have conducted numerous studies to investigate the application of AI in transportation. Along the journey to strive for better solutions, it is essential to conduct periodical reviews on state-of-the-art

achievements to discover the prevailing research trends in road safety. Moreover, this helps to identify existing research gaps and propose future directions for AI solutions to enhance road safety. There is a comprehensive review that discusses the existing deep learning-based applications in the transportation domain, covering the representation of traffic networks, traffic flow prediction, travel demand prediction, traffic signal control, vehicle detection, autonomous driving, traffic incident analysis, and driver behaviors analysis (Nguyen et al., 2018). The study analyzed the application of deep learning algorithms such as convolutional neural networks (CNN), recurrent neural networks (RNN), and deep belief networks (DBN) in the transportation domain. The potential of deep learning algorithms in the transportation domain is highlighted and future research is recommended to investigate the effects of spatial-temporal features on traffic incidents. However, there is limited discussion on key variables that affect the performance of existing deep learning-based road safety solutions.

Apart from the broad transportation domain, Silva et al. (2020) attempt to survey the ML-based application in a more specific context, which is road safety modeling. The study focuses on the solutions to model crash severity and frequency. It reveals that the common ML algorithms utilized in this context are k-nearest neighbors (KNN), decision trees (DT), evolutionary algorithms (EA), support vector machines (SVM), and artificial neural networks (ANN). Furthermore, this study also identifies the common variables used for road safety modeling, including road and weather conditions, vehicle characteristics, driver's behavior, and historical crash data. The review thoroughly discusses ML-based crash severity and crash frequency modeling. Nevertheless, the scope can be expanded to include other tasks such as crash risk and occurrence. Moreover, it is worthwhile to outline a standard framework based on the prevailing trends discovered throughout the review. This can serve as guidelines for future research on ML-based road safety modeling.

In terms of the data sources, the review by Gutierrez-Osorio and Pedraza (2020) points out that the common sources being utilized in RTA analysis and prediction are social media data, GPS data, and weather data. The study also highlights that there is limited research that considers variables such as traffic flow, human mobility, and extraordinary events. It suggests future research to include heterogeneous data sources to boost the performance of the RTA analysis and prediction models. Over the years, there are studies that incorporate heterogeneous data such as traffic occupancy and annual average daily traffic (AADT) in road safety modeling (Alqatawna et al., 2021; Amiri et al., 2020; Cai et al., 2020; Jiang et al., 2020; J. Wang et al., 2022; Yang et al., 2022). Hence, there is a need to review and evaluate the effectiveness of heterogeneous data for ML-based RTA prediction. Marcillo et al. (2022) have performed this and recommended a direction to address the issues of high dimensionality and information imbalance among heterogeneous data. However, this is limited to the studies published before the year 2021 and further updates are necessary to identify the recent progress.

To date, there is still limited systematic review of ML-based RTA prediction with non-visual data. While some existing reviews discuss the application of ML in road safety modeling, it is vital to include recent studies to keep track of the progress and identify potential research directions in this field. With this motivation, this work is performed to critically review the recent solutions related to ML-based RTA prediction with non-visual data. It is expected to contribute to the following:

- Identifies the common sources of non-visual data used for ML-based RTA prediction and determines their limitations.
- Covers a wider range of tasks comprising crash severity, frequency, risk, and occurrence modeling for a more thorough understanding of the latest trends under these categories.
- Compares the performance of various ML algorithms and identifies the existing gaps of ML-based RTA prediction with non-visual data.

- Determines the potential directions for future research based on the overall limitations of ML-based RTA prediction observed from recent studies.

3. Methodology

Presently, there are various systematic reviews of state-of-the-art ML-based road safety modeling technologies. However, there is little to no review that evaluates the current progress of ML-based RTA prediction which specifically involves the non-visual data. Thus, the discovery is performed through this work by adopting Kitchenham's guidelines for systematic review (Kitchenham, 2004). The guidelines are modified into a workflow comprised of three stages: (1) review planning, (2) search and filtration, and (3) analysis and reporting. The review planning stage involves defining the research problem and formulating research questions. With the research questions, the process continues with the search and filtration stage which involves the activities of searching and selecting studies relevant to ML-based RTA prediction. These include identifying academic databases, determining search keywords, and formulating inclusion-exclusion criteria for quality assessment. After collecting all the selected studies, the analysis and reporting stage is initiated to extract useful data from the studies, which is then analyzed to answer the research questions.

3.1. Review planning

The ML-based RTA prediction with non-visual data is deemed more effective for the LMICs in line with their current traffic management system and infrastructure which might not be as advanced as the high-income countries. For this reason, it is essential to keep track of the existing research progress on ML-based RTA prediction with non-visual data. This is significant for identifying the current trends and gaps, enabling recommendations of future research directions to support the LMICs in achieving the goal of reducing RTA fatalities. With this motivation, this review is conducted and limits the scope to focus on ML-based RTA prediction with non-visual data. As the outcome of the review planning stage, several research questions (RQs) are formed as guidelines to discuss the findings in this work.

- RQ1. What are the limitations of data sources used in the existing studies?
- RQ2. How good is the performance of the ML algorithms implemented in the existing studies?
- RQ3. How do the strengths and limitations affect the selection of ML algorithms in the existing studies?
- RQ4. What is the future direction for ML-based RTA prediction with non-visual data?

3.2. Search and filtration

After forming the RQs, the process moves into the search and filtration stage to collect relevant literature through the search and filtration activities. The search and filtration strategies are developed to obtain a list of relevant studies for data extraction and analysis in the analysis and reporting stage.

3.2.1. Search strategy

The search process involves a series of manual operations such as identifying academic databases, listing potential search keywords, and conducting the search. Firstly, the searches are conducted on academic databases from ACM, Elsevier, IEEE Xplore, MDPI, Springer, Taylor and Francis, and Wiley. These databases are selected because they are expected to encompass publications related to science, technology, engineering, and transportation. The search is also performed through Scopus to source relevant publications not included in the above-mentioned academic databases. To keep track of the recent research

progress, the search is also limited to include only the studies published from 2018 onwards (within the last 5 years). In addition, a cross-reference search is also conducted to identify additional literature cited in the collected studies. These studies are eventually added to the collection and brought for the filtration process.

In this work, the literature search is performed by searching the identified phrases from the title, abstract, and keywords. The phrases are formed through the combination of three categories, which are the subject ("road traffic accident"), task ("prediction"), and technique ("machine learning"). These categories are combined with the "AND" operator as the search phrase. In addition, this work also identified the synonyms for each category as in Table 1. They are included by using the "OR" operator to obtain a more inclusive search result. Moreover, the wildcard operator, "*" is applied to cover all the variations. For instance, "predict*" helps to represent all variations of the word "predict" such as "predict", "predicts", "predicting", "predicted", and "prediction". Eventually, the string supplied to the advanced search tool has the following format:

("road traffic accident" OR "roadway accident" OR "highway accident" OR "road crash" OR "road collision" OR "traffic mishap" OR "vehicular accident" OR "road transport accident" OR "mixed traffic accident" OR "car accident" OR "fatal accident") AND ("predict*" OR "forecast*" OR "classif*" OR "analy*" OR "categor*" OR "model*") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "neural networks" OR "supervised learning" OR "unsupervised learning" OR "learning-based")

3.2.2. Filtration strategy

After performing the literature search from the academic databases, the next step is to filter and finalize the relevant studies for this review. The filtration process starts with a preliminary screening of the title and abstract to eliminate the studies that are not related to the ML-based RTA prediction. Then, the process continues with the de-duplication to remove the duplicated studies so that each study is only evaluated once. Finally, the eligibility assessment is performed to assess the quality of the studies based on the inclusion and exclusion criteria. The studies are expected to satisfy all the inclusion criteria for them to be shortlisted for evaluation. Moreover, a study is rejected when any of the exclusion criteria is detected. For this work, the inclusion and exclusion criteria are determined as follows.

Inclusion Criteria:

1. The study is published in scientific, technology, engineering, and transportation journals or conferences.
2. The study is published as a peer-reviewed research paper.
3. The study developed and investigated ML-based models for RTA prediction.
4. Performance metrics are used to evaluate the performance of the proposed models.
5. Benchmarking is done to compare the performance of the proposed model with other baseline models.

Exclusion criteria:

Table 1
Keywords for literature search in this work.

Category	Keyword
Subject	road traffic accident, roadway accident, highway accident, road crash, road collision, traffic mishap, vehicular accident, road transport accident, mixed traffic accident, car accident, fatal accident
Task	prediction, forecast, classification, analysis, categorization, modeling
Technique	machine learning, deep learning, artificial intelligence, neural networks, supervised learning, unsupervised learning, learning-based

1. The work is published in journals or conferences unrelated to the topic.
2. The work does not propose an ML-based model for RTA prediction.
3. The work does not involve non-visual data for RTA prediction.
4. There is no full text available or inaccessible.

3.3. Analysis and reporting

In the analysis and reporting stage, data extraction is executed to extract useful information from the shortlisted studies. The information is used as the baseline to answer the research questions formulated during the review planning stage. In this context, it includes the type of non-visual data and ML algorithms, followed by the key limitations observed in the shortlisted studies. Then, analysis is made on the extracted data to synthesize information such as the common categories of non-visual data usually involved in RTA prediction and the trend of ML algorithms application in recent studies. This review also attempts to compare the performance of the RTA prediction models and discuss the existing research gaps to determine the potential research direction for ML-based RTA prediction with non-visual data.

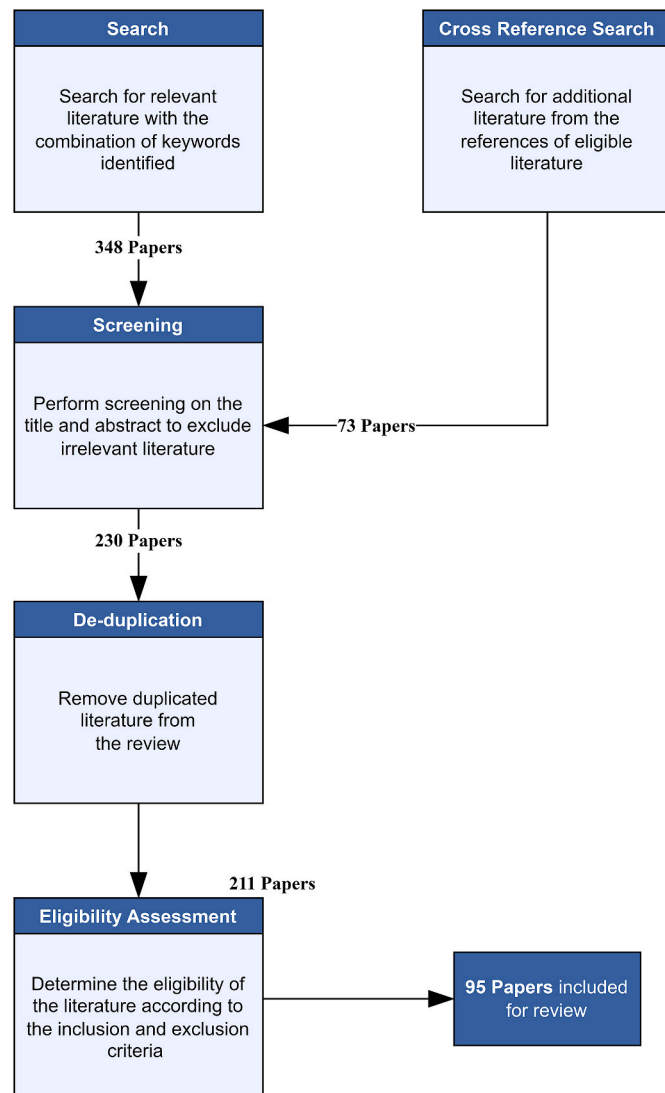


Fig. 1. Literature search and filtration for this work.

4. Findings and discussion

The literature search process (see Fig. 1) has collected 421 papers, and 210 papers are excluded through screening and de-duplication. This resulted in 211 papers going into the subsequent eligibility assessment based on the inclusion and exclusion criteria, as discussed in Section 3.2. Subsequently, this assessment shortlisted 95 studies that are relevant to non-visual ML-based RTA prediction, as summarized in Table 2. Based on Fig. 2, they comprise 70 journal articles (74%), 19 proceeding articles (20%), and 6 book chapter articles (6%). Moreover, the journal articles are found to have considerable quality with 62 studies (88%) from Quartile 1 and Quartile 2 journals. Apart from that, an interesting trend is observed based on the distribution of studies across the six WHO regions. The Americas and European regions which have the most studies conducted have lower average RTA fatality rates at 15.33 and 7.40 (deaths per 100,000 population) respectively. Furthermore, the literature indicates a lower focus on the African and Eastern Mediterranean regions, which have more LMICs.

The shortlisted studies are discovered to focus on different RTA prediction tasks and are grouped into four major modeling categories comprising crash risk modeling (CRM), crash occurrence modeling (COM), crash severity modeling (CSM), and crash frequency modeling (CFM). These categories are formed on top of the characteristics of RTA prediction defined as follows.

- **CRM:** An approach to model the probability for a crash to occur at a specific area or road segment based on the RTA features obtained within a dedicated period. In other words, this modeling category is often a regression task that estimates the road safety level at different road segments.
- **COM:** Modeling the crash occurrence is related to the task of classifying the inputs into the crash or non-crash category. It is a classification task that identifies the case of RTA occurrence based on the inputs, while CRM predicts the risk of future crashes.
- **CSM:** Development of a prediction model to classify the severity level of an RTA based on the RTA feature provided. The classification output is often categorized into non-injury, slight injury, serious injury and fatality.
- **CFM:** The main task under CFM is to estimate the number of RTA on a road segment within a specific period. Its main objective is to identify the hotspots of RTA based on the predicted frequency so that appropriate mitigation plans can be formulated.

Out of the defined categories, 59 studies (62%) fall into the CSM category, where various ML algorithms are applied to predict the severity level of RTAs. This is performed based on a series of non-visual data describing the RTA. Despite the multiclass classification of RTA severity (non-injury, injuries, and fatalities), some studies focused on a more specific category such as fatal or non-fatal (Siddik et al., 2021), pedestrian fatality or non-pedestrian fatality (Tao et al., 2022) the extent of damage (Vadhwani and Thakor, 2022) and the level of traffic delay (Z. Wang et al., 2022). Following that, COM is the second most investigated category, with 16 studies (17%) focusing on performing binary classification of RTA occurrence (crash or non-crash). For this category, the data from hardware and sensors such as loop detectors are beneficial due to their ability to provide real-time information along the road (Parsa et al., 2019; Theofilatos et al., 2019). Next, there are 12 studies (13%) from the CFM category that perform regression tasks to forecast the frequency of RTA for a specific time. Another perspective of CFM can be the estimation of RTA casualties' frequency (Gu et al., 2018; Z. Zhang et al., 2020). These predictions are essential to illustrate the expected frequency of RTA and casualties to ensure the authorities' preparedness during such events. The CRM is found to be the least popular category among the selected studies with 8 studies (8%) that predicted RTA risk based on the real-time traffic condition information. Historical RTA records are the common data being used to estimate the RTA risk.

Table 2

Summary of the literature shortlisted for ML-based RTA prediction with non-visual data.

Code	Author	Country	Modeling Category	Task Type	Contributions	Key Factors Observed
S01	Abdulrashid et al. (2024)	US	CSM	C	Formulated an explainable AI framework that uses ML algorithms to predict RTA severity in the US, with added functionality to understand and identify the main risk factors.	B, C, H
S02	Aboulola (2024)	New Zealand	CSM	C	Demonstrated the use of transfer learning model to predict RTA severity for the case of New Zealand and applied the SHAP algorithm to analyze the significance of various RTA features.	B, E
S03	Balawi and Tenekeci (2024)	UK	CFM	R	Benchmarked the performance of different statistical ML algorithms to estimate RTA frequency on London's A-roads and identify the significant factors.	B, C, F, H
S04	Guo et al. (2024)	US	CRM	R	Evaluated and quantified pedestrian exposure to RTA based on multi-dimensional factors and developed an ensemble deep learning model to capture spatiotemporal variations of features to estimate RTA risk.	D, F
S05	Khan et al. (2024)	US	CSM	C	Adopted the DeepInsight technique for numerical-to-image conversion and trained a transfer learning model to predict the severity of RTAs involving heavy trucks at mountainous Interstate-80 (I-80) in Wyoming.	B, C, E, H
S06	Mo et al. (2024)	US	COM	C	Analyzed the contributing factors of dynamic short-term crash risk and developed a hybrid model to predict RTA occurrence at mainline toll plazas in the US.	C, E, G
S07	Raul et al. (2024)	Social media	CSM	C	Proposed a weighted ensemble of self-training semi-supervised learning approaches to classify RTA severity based on the data extracted from RTA-related tweets.	
S08	Shao et al. (2024)	China	CSM	C	Integrated XGBoost with natural language processing (NLP) to predict RTA severity while offering a comprehensive analysis of textual contents in the crash reports for better understanding of the RTAs.	B, E, H
S09	Hasan et al. (2023)	US	CSM	C	Investigated various ML algorithms for predicting the severity of RTAs involving young drivers in New Jersey, supported by sensitivity analysis and SHAP values to analyze the contributing factors.	C, E
S10	Kashifi et al. (2023)	France	COM	C	Introduced the Deep Spatiotemporal Hybrid Network (DSHN), an innovative model that utilizes spatiotemporal information for short-term crash occurrence prediction.	G, H
S11	Sattar et al. (2023)	UK	CSM	C	Discovered the feasibility of TabNet to classify RTA severity in the UK with high interpretability.	A, D, H
S12	Zhang et al. (2023)	UK	CSM	C	Innovatively focused on single-bicycle crashes in the UK by applying Ordered Forest (ORF) to predict and analyze the RTA severity for gaining insights into potential bicycle safety countermeasures.	A, B, D, E, F
S13	Danesh et al. (2022)	Iran	CSM	C	Introduced a new technique known as under sampling based on clustering using metaheuristic (USBCM) approach to balance the RTA dataset for boosted performance of RTA severity prediction.	A, C, E, H
S14	Dia et al. (2022)	Senegal	CSM	C	Advanced the RTA severity prediction solution in Senegal from traditional statistics towards ML-based models.	
S15	Emu et al. (2022)	Canada	CSM	C	Investigated the applicability of training the CNN-based model with a large dataset to classify Canadian crashes into non-fatal or fatal categories.	
S16	Infante et al. (2022)	Portugal	CSM	C	Contributed to a comparative study between statistical regression and conventional ML to suggest the best model for predicting RTA severity in Portugal.	A, B, C, D, E, F, H
S17	J. Wang et al. (2022)	US	COM	C	Conducted a comparative investigation between deep learning (CNN) and statistical approaches (BLR) to classify crash occurrence at freeway work zones in the US.	G
S18	Mohanty et al. (2022)	India	CSM	C	Developed ML-based models to predict RTA severity in India with a distinct focus on the role of perpetrators and victims in RTAs.	A, H
S19	Nassiri et al. (2022)	Iran	CFM	R	Contributed to the time-series analysis of RTAs that has not been evaluated in Iran by using random forest (RF) for short-term crash frequency prediction.	
S20	Panda et al. (2022)	India	CSM	C	Advanced the ML-based RTA severity prediction solutions in India by integrating explainable AI techniques to interpret model predictions.	A, B, H
S21	Shi et al. (2022)	US	COM	C	Introduced a novel deep learning-based feature selection technique to capture the characteristics from noisy high-frequency kinematics data to improve the performance of the ML model in detecting crash and near-crash events.	
S22	Tao et al. (2022)	Australia	CSM	C	Attempted to improve the prediction and analysis of pedestrian deaths due to RTAs in Australia by utilizing the Bayesian neural network (BNN) that addresses the overfitting and generalization issues of typical neural networks (NN).	A, B, C
S23	Vadhwani and Thakor (2022)	US	CSM	C	Discovered the use of ML algorithms for classifying RTA severity in the US from the perspective of the extent of damage experienced by the involved vehicles.	
S24	Yang et al. (2022)	China	CRM, COM	C, R	Introduced the use of Reinforcement Learning Tree (RLF) for multitask RTA prediction in China, which are the crash risk and occurrence.	G
S25	Z. Wang et al. (2022)	US	CSM	C	Conducted a comparative study to evaluate performance of both the RF and light gradient boosting machine (LightGBM) to predict the RTA severity in the US in terms of the level of delay.	C, D, F
S26	Alnami et al. (2021)	US	CSM	C	Attempted to support the emergency response of RTAs in the US by proposing an ML-based solution to classify RTA severity so that optimal allocation of required emergency vehicles can be made.	C, E, F
S27	Alqatawna et al. (2021)	Spain	CFM	R	Advanced the estimation of RTA frequency on Spain's highways and the comparative analysis proposed that ANN can be a potential alternative for the traditional statistical approach.	G

(continued on next page)

Table 2 (continued)

Code	Author	Country	Modeling Category	Task Type	Contributions	Key Factors Observed
S28	Ashikuzzaman et al. (2021)	UK	CSM	C	Introduced the integration of Particle Swarm Optimization (PSO) for hyperparameters tuning of ANN model to boost performance in RTA severity prediction.	
S29	Bedane et al. (2021)	Ethiopia	CSM	C	Contributed to an RTA dataset generated from police reports in Addis Ababa, Ethiopia for investigating ML-based RTA severity prediction and factor analysis.	B, C, E
S30	Beryl Princess et al. (2021)	India	CSM	C	Contributed to addressing the class imbalance issue of the RTA dataset with the SMOTE technique and evaluated its effectiveness with several ML models to predict RTA severity in India.	
S31	Chen (2021)	US	CSM	C	Investigated the applicability of CNN to perform RTA severity predictions, with a distinct focus on the urban road network in Chicago.	
S32	Ghasedi et al. (2021)	Iran	CSM	C	Conducted a case study for Iran with the logit model and factor analysis to determine key factors and evaluate several ML algorithms in classifying RTA severity.	B, C, H
S33	Ijaz et al. (2021)	Pakistan	CSM	C	Advanced the ML-based solutions for RTA severity prediction in Pakistan by specifically focusing on the RTAs involving three-wheeled motorized rickshaws.	B, C, E, F
S34	Rahim and Hassan (2021)	US	CSM	C	Suggested a potential application of CNN for classifying RTA severity with the numeric-to-image conversion technique to transform non-visual RTA data into images.	A, B
S35	Reichenbach et al. (2021)	Germany	COM	C	Introduced the use of emergency braking data as one of the features to consider by the ML model for detecting RTA incidents in Germany.	C
S36	Shetty and Liu (2021)	US	CSM	C	Conducted a comparative study to benchmark the statistical regression and conventional ML techniques and identify the best model for predicting traffic delayed time as RTA severity in the US.	
S37	Siddik et al. (2021)	Bangladesh	CSM	C	Proposed web scraping as an alternative source for collecting RTA dataset and examined its applicability for training ML-based models for predicting RTA severity in terms of the presence of death.	
S38	Zhao and Rao (2021)	US	CFM	R	Introduced the concept of multi-factor model that considers RTA factors such as RTA severity, weather conditions, and external geographic environment to improve the accuracy of predicting RTA frequency.	
S39	Zhu et al. (2021)	US	CSM	C	Attempted to improve ML-based RTA severity prediction by introducing an ordinal ML classification approach that analyzes the severity levels as ordinal data.	A, B, H
S40	Amiri et al. (2020)	US	CSM	C	Introduced an ML-based model to predict and analyze severity of RTAs in the US at a more refined scope focusing on run-off-road (ROR) fixed object crashes involving elderly drivers.	C
S41	Assi (2020)	Australia	CSM	C	Incorporated the principal component analysis (PCA) for feature reduction and evaluated its effectiveness for optimizing RTA severity prediction for the case of Australia.	A, B, C, E, H
S42	Assi et al. (2020)	UK	CSM	C	Applied the Fuzzy C-Means (FCM) algorithm to cluster the RTA data samples and evaluated the effects of the number of clusters on the accuracy of ML models to classify RTA severity.	
S43	C. Zhang et al. (2020)	US	CSM	C	Introduced the use of PSO to tune the weights of ANN so that the best weights are obtained for better performance of RTA severity prediction.	A, B, D, E, F, H
S44	Cai et al. (2020)	US	COM	C	Introduced the use of a generative network, the deep convolutional generative adversarial network (DCGAN) to balance the RTA dataset so that the ML-based model is better trained to detect RTA occurrence.	G
S45	Donchenko et al. (2020)	Russia	CSM	C	Contributed to a case study of ML-based RTA severity prediction in Russia while limited studies have investigated it for this region.	B, C, E, F, H
S46	Elamrani Abou Elassad et al. (2020)	Simulation	COM	C	Adopted driving simulation data balanced with Synthetic Minority Oversampling Technique (SMOTE) to train a model for crash occurrence prediction and key factors analysis.	
S47	Fiorentini and Losa (2020)	UK	CSM	C	Introduced the random undersampling of majority class (RUMC) technique to balance the RTA datasets so that the performance of ML models to predict RTA severity is improved, particularly for the minority severity class.	A, E, F, H
S48	Hadjidimitriou et al. (2020)	US	CSM	C	Developed a solution to determine the need for emergency calls for RTAs involving two-wheelers based on the predicted severity by ML models.	A, B, H
S49	Haruna et al. (2020)	Australia	CSM	C	Introduced a unique solution for selecting significant features to improve the predictive performance and reduce the training time of the deep learning-based RTA severity prediction model.	
S50	Huang et al. (2020)	US	COM	C	Investigated the applicability of deep learning-based RTA occurrence prediction by using the traffic data collected from the sensors installed on highways in the US.	B, C, G
S51	Jiang et al. (2020)	US	COM	C	Proposed the LSTMDTR framework, an innovative solution based on Long Short Term Memory (LSTM) to detect RTA occurrence at different temporal resolutions.	A, E, G, H
S52	Katanalp and Eren (2020)	Türkiye	CSM	C	Investigated the effectiveness of integrating fuzzy logic (FL) with DT to improve RTA severity predictions, with a distinct focus on bicycle-vehicle crashes.	A, B, E, H
S53	Lee et al. (2020)	Korea	CSM	C	Conducted a comparative study to benchmark the performance of various ML algorithms for RTA severity predictions in Seoul city, specifically focusing on the RTAs that occurred during rainy seasons.	B, C, E
S54	Li et al. (2020)	US	CRM	R	Proposed an innovative solution that combines the LSTM and CNN so that both the long-term dependencies and time-invariant features can be captured to better estimate RTA risks in real-time.	C, G, H
S55	Lin et al. (2020)	US	CSM	C	Attempted to focus on the RTAs involving teen drivers in the US by applying ML algorithms to predict the severity and identify key factors that contribute to it.	A, B, C, D, E

(continued on next page)

Table 2 (continued)

Code	Author	Country	Modeling Category	Task Type	Contributions	Key Factors Observed
S56	Panicker and Ramadurai (2020)	India	CSM	C	Specifically focused on the severity prediction of RTAs involving two-wheelers at mid-block road sections with ML algorithms.	A, B, C, H
S57	Rezapour et al. (2020)	US	CSM	C	Examined the parametric and non-parametric ML methods to analyze and predict RTA severity in the US, specifically motorcycle at-fault crashes.	B, E
S58	Sarkar and Sarkar (2020)	India	CSM	C	Proposed the ensemble ranking methods comprising information gain, gain ratio, and symmetric uncertainty to select significant features for improving the ML-based RTA severity prediction in India.	A, B, E, G, H
S59	Singh et al. (2020)	India	CFM	R	Promoted the use of deep learning to develop models for predicting RTA frequency on India's highways.	E, G
S60	Tariq et al. (2020)	Yemen	CSM	C	Addressed the lack of an RTA dataset in Yemen by collecting data from the hospitals' emergency departments and used the dataset to build ML-based models for predicting RTA severity.	A, B, E
S61	Wahab and Jiang (2020)	Ghana	CSM	C	Conducted a case study for the RTAs in Ghana to discover the applicability of training ML models to classify the severity of crashes involving motorcycles.	A, D, E, F, H
S62	Wu and Wang (2020)	China	CFM	R	Performed a comparative analysis of performance between conventional ML and deep learning algorithms to identify the best model for estimating RTA frequency in China.	G, H, I
S63	Yassin and Pooja (2020)	Ethiopia	CSM	C	Combined the K-Means and RF as a new hybrid approach to improve model accuracy for classifying RTA severity based on the Addis Ababa RTA dataset.	A, B, C, D, F, H
S64	Z. Zhang et al. (2020)	China	CFM	R	Proposed a novel hybrid model that consists of LSTM for extracting time dependencies and learned by gradient-boosted regression trees (GBRT) to forecast the number of casualties as the RTA risk indicator.	
S65	Zhao et al. (2020)	UK	CRM	R	Developed a hybrid model with CNN and RF to estimate RTA risks of different regions in the UK based on data collected from mobile edge computing.	
S66	Almamlook et al. (2019)	US	CSM	C	Developed an ML-based model that can be used by the Michigan Traffic Agencies to classify RTA severity and analyze influential factors.	
S67	Chakraborty et al. (2019)	India	CFM	R	Developed an ANN-based model for RTA frequency estimation, specifically focusing on pedestrian fatal crashes at intersections in India.	B, C, G, H
S68	Kumeda et al. (2019)	UK	CSM	C	Contributed literature for the ML-based RTA severity prediction in the UK with a comparative study to evaluate and identify the best performing model among the six ML algorithms.	A, B, C, E
S69	Labib et al. (2019)	Bangladesh	CSM	C	Contributed to a case study in Bangladesh for the application of ML algorithms to analyze RTA characteristics and predict severity levels.	
S70	Mokhtarmousavi et al. (2019)	US	CSM	C	Developed an improved SVM model to predict and analyze the RTA severity that specifically occurred at work zones in the US.	A, D, E, F
S71	Mokoatle et al. (2019)	South Africa	CSM	C	Advanced the ML-based RTA severity prediction in South Africa with additional considerations on features related to the distance to the closest places of interest.	A, D, H
S72	Parsa et al. (2019)	US	COM	C	Performed a case study with the dataset from Eisenhower expressway, Chicago and compared the performance of SVM and probabilistic neural network (PNN) to detect RTA occurrence.	G, H
S73	Theofilatos et al. (2019)	Greece	COM	C	Conducted a comparative analysis between conventional ML and deep learning using the RTA dataset in Greece to determine the best model for detecting RTA occurrence.	C, G
S74	Wahab and Jiang (2019)	Ghana	CSM	C	Investigated the development of ML-based solutions to predict RTA severity in Ghana, with a refined scope of motorcycle-involved crashes.	A, D, E, F
S75	Wang and Kim (2019)	US	CSM	C	Conducted a study to compare the performance of discrete choice and tree-based models for classifying RTA severity and identify the significant contributing factors.	A, B, C, E, H
S76	Xi et al. (2019)	China	CSM	C	Introduced the use of rough set theory to optimize the performance of SVM and evaluated its effectiveness for classifying the RTA severity in China.	
S77	Yahaya et al. (2019)	Ghana	CSM	C	Proposed a framework for improving the ML-based RTA severity prediction in Ghana with data balancing and feature selection techniques.	A, D, E
S78	Zhang et al. (2019a)	US	CRM	R	Introduced RiskCast, a novel solution that combines both social sensing and open city data to forecast RTA risks in the US.	
S79	Zhang et al. (2019b)	US	CRM	R	Proposed DeepRisk, an improved framework that utilizes deep transfer learning techniques for developing a model to estimate RTA risk based on social media data.	
S80	Zhao et al. (2019)	UK	CRM	R	Attempted to optimize model performance to predict RTA risk in the UK with a hybrid of factorization machine (FM) and CNN.	
S81	Zheng et al. (2019)	UK	CSM	C	Introduced a novel technique to convert the tabular RTA data into gray images, which are then used for training the CNN model to predict RTA severity.	
S82	Al-Radaideh and Daoud (2018)	UK	CSM	C	Applied the PART algorithm, a data mining technique to generate the decision rules for key factors analysis and incorporated with ML algorithms to predict RTA severity.	A, C, D
S83	Bahiru et al. (2018)	UK	CSM	C	Applied various conventional ML algorithms to predict the severity of RTAs in the UK and used them to determine the key contributing factors.	C, E, F
S84	Basso et al. (2018)	Chile	COM	C	Develop a model for detecting RTA occurrence on urban expressways in Chile based on the real-time disaggregated data collected from free-flow toll gates.	G
S85	Dogru and Subasi (2018)	Simulation	COM	C	Explored the feasibility of using simulation data collected from vehicular ad-hoc networks (VANETs) to train an ML-based model for RTA occurrence detection.	
S86	Dong et al. (2018)	US	CFM	R	Improved the performance of the deep learning model to estimate crash frequency by integrating a multivariate regression layer to resolve the issue of unobserved heterogeneity.	

(continued on next page)

Table 2 (continued)

Code	Author	Country	Modeling Category	Task Type	Contributions	Key Factors Observed
S87	Gu et al. (2018)	China	CFM	R	Proposed an improved PSO algorithm known as the particle swarm optimization with mutation (PSOM) for tuning the SVM to improve the estimation of RTA fatality frequency in China.	A, E, I
S88	Hasheminejad et al. (2018)	Iran	CSM	C	Enhanced the ML-based solution in Iran with a novel rule-based genetic algorithm (GA) that performs the clustering process before classifying RTA severity.	A, B, C, F, H
S89	J. Zhang et al. (2018)	US	CSM	C	Performed performance benchmarking between statistical regression and conventional ML algorithms to classify RTA severity based on the Florida dataset.	A, C, E
S90	Jeong et al. (2018)	US	CSM	C	Trained various ML-based models with the Michigan Traffic Crash Facts dataset to identify the best model for classifying the severity of motor-vehicle crashes.	
S91	Liu et al. (2018)	US	CFM	R	Attempted to perform a categorical prediction of RTA risk in the US by utilizing ML algorithms to classify the data into high-, medium-, and low-frequency categories.	H
S92	Meng et al. (2018)	China	COM	C	Proposed an ML-based solution that performs a binary classification of RTA occurrence in China to address the issue of the limited number of RTAs per hour.	C, E, F, H
S93	Ren et al. (2018)	China	CRM	R	Proposed the use of spatiotemporal correlation patterns to forecast RTA risks with deep learning algorithms.	
S94	Singh et al. (2018)	India	CFM	R	Evaluated the use of statistical regression and conventional ML algorithms to forecast RTA frequency on India's highways.	E, G
S95	Z. Zhang et al. (2018)	US	COM	C	Explored a novel approach to detect RTA occurrence through deep learning algorithms and extracted social media data (tweets content) as inputs.	

Modeling Category: CFM = Crash frequency modeling; COM = Crash occurrence modeling; CRM = Crash risk modeling; CSM = Crash severity modeling.

Task Type: C = Classification; R = Regression.

Key Factors Observed: A = Accident information; B = Road users (drivers or victims) profile; C = Environmental conditions; D = Geographical (location) information; E = Road characteristics and conditions; F = Time related information; G = Traffic conditions; H = Vehicle characteristics and conditions; I = Others.

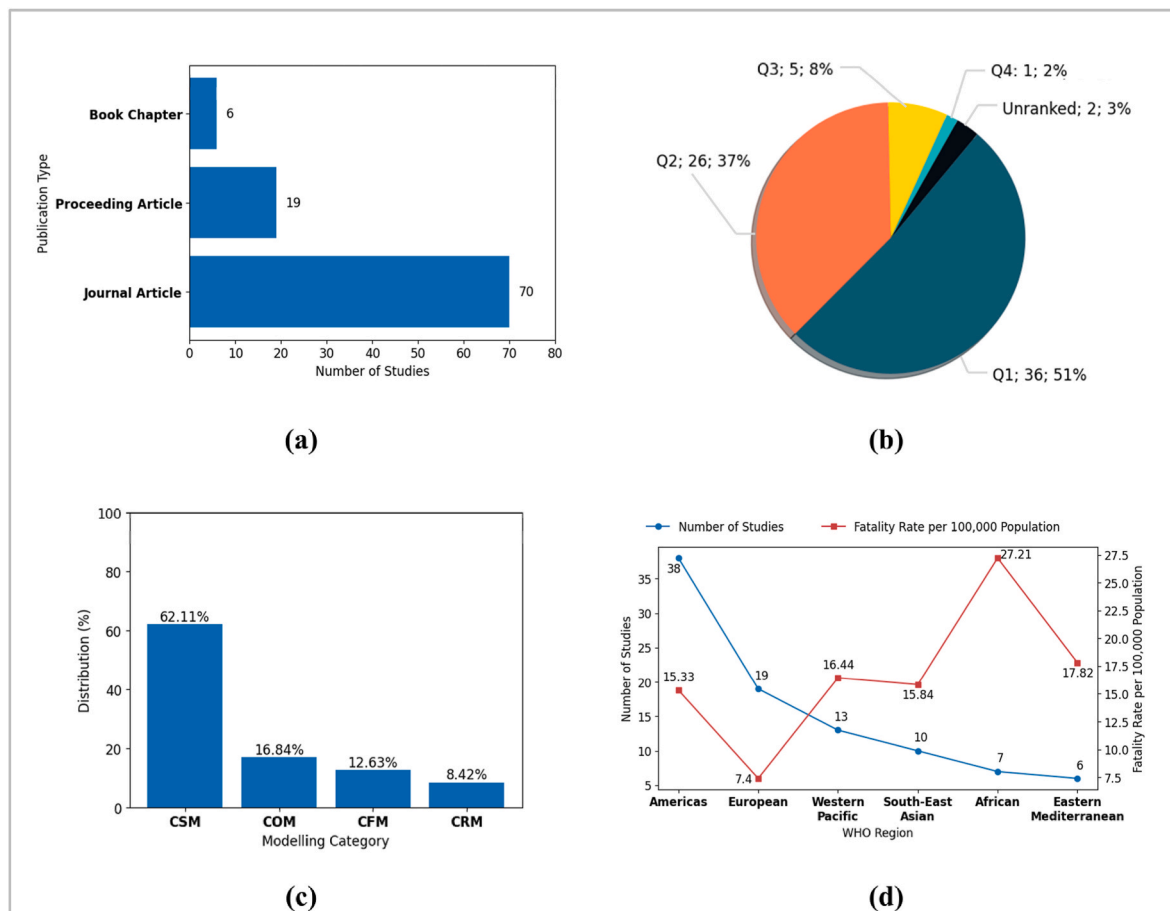


Fig. 2. Distribution of ML-based RTA prediction studies in terms of (a) publication type, (b) journal quartile ranking, (c) modeling category, and (d) WHO region.

However, there exist some exceptional cases that utilize social media data such as historical Twitter content for RTA risk prediction, highlighted through solutions such as RiskCast and DeepRisk (Zhang et al., 2019a, 2019b).

4.1. Road traffic accident data sources

This section discusses the sources of non-visual RTA datasets, often available in tabular form, used in the selected studies for ML-based RTA prediction. The strengths and limitations of the data sources are then identified to gain insights into the factors that affect the selection of data sources for future research. This is vital because an appropriate RTA dataset is considered the key to a high-quality investigation of ML-based RTA prediction. The findings from the selected studies showed that non-visual RTA data can be sourced from a range of diverse sources as illustrated in Fig. 3, including the government department (50%), open-access database (34%), sensor (8%), social media (4%) and other (4%). Each of the sources has its strengths and limitations and is critically analyzed in Section 4.1.6.

4.1.1. Government department

Government departments, including the police service (Bedane et al., 2021; Chakraborty et al., 2019; Das et al., 2020; Mokoatle et al., 2019; Panicker and Ramadurai, 2020; Sarkar and Sarkar, 2020; Singh et al., 2018; Yassin and Pooja, 2020) and the transportation departments or agencies (Almamlook et al., 2019; Danesh et al., 2022; Dong et al., 2018; Huang et al., 2020; Ijaz et al., 2021; Infante et al., 2022; Lee et al., 2020; Meng et al., 2018; Parsa et al., 2019; Ren et al., 2018; Zheng et al., 2019) are entrusted with tasks related to road safety and management, including the collection and maintenance of the historical RTA records. This makes them the most popular sources for collecting datasets comprising historical RTA records. Additionally, some studies also sought data from other departments for specific information. For instance, Dia et al. (2022) adopted the data from a collaborative survey conducted by the Ministry of Economy and insurance companies in Senegal to train models for RTA severity predictions. In some studies, meteorological agencies such as the Korea Meteorological Administration (Lee et al., 2020) and the National Oceanic Atmospheric Administration (Li et al., 2020) are used to obtain weather-related data.

Apart from that, some universities and research institutes have taken the initiative to develop their own databases of RTA data for research purposes, such as the University of Connecticut (Zhao and Rao, 2021), Building and Road Research Institute of the Council for Scientific and Industrial Research (BRRI-CSIR) in Ghana (Wahab and Jiang, 2019, 2020; Yahaya et al., 2019), Accident Research Institute at Bangladesh University of Engineering and Technology (Labib et al., 2019), Traffic Accident Research Institute of China Automotive Technology and Research Center (Jianfeng et al., 2019), and Department of Transportation Planning and Engineering of the National Technical University

of Athens (Theofilatos et al., 2019). These databases served as a valuable platform to access reliable non-visual RTA datasets, supporting the investigation of various RTA prediction tasks. In specific circumstances, such as when political issues impede access to historical RTA records from police services, the potential of alternative sources, such as hospitals is explored. A case in point is the work from Tariq et al. (2020), where RTA data is collected from the emergency departments at Sciences and Technology University Hospital (STUH) and Al-Gumhouri General Hospital (AGH) due to the limited access to road traffic police database in Yemen. The researchers have opted for secondary sources to obtain RTA-related data by interviewing the hospital's injured patients.

4.1.2. Open-access database

Another common source of non-visual RTA datasets in recent studies is the open-access database. It is a website or platform containing a wide range of datasets that are freely available to the public for various purposes, such as research, analysis, and policymaking. In the context of RTA, existing studies mostly used open-access RTA datasets published through the official website or open data portal of related departments, such as the police, transportation, and Bureau of Statistics. Furthermore, non-governmental open-access platforms also serve as valuable sources for collecting RTA datasets. For example, Kaggle is a non-governmental platform that hosts and shares various datasets, making them available to the public for ML and data analysis. Shetty and Liu (2021) have used the "US Accidents" dataset obtained from Kaggle to train and assess the performance of various models, including regression, tree-based, ensemble learning, and gradient-boosting models for predicting the severity of RTA. Moreover, Emu et al. (2022) utilized the "Canadian Car Accidents 1994–2014" dataset from Kaggle to explore the applicability of CNN in classifying motor vehicle crashes in Canada into categories such as fatal and non-fatal RTA.

4.1.3. Sensor

In the context of RTA predictions, the use of sensors can be highly advantageous to collect real-time data related to road conditions or traffic volume. The real-time data collected through the sensors can be supplied to the RTA prediction models for different purposes, such as the prediction of risk, occurrence, and frequency of RTA, as discovered through the existing studies. For instance, traffic flow sensors such as inductive loops (Theofilatos et al., 2019), loop detectors (Parsa et al., 2019), and BlueMAC (Li et al., 2020) are used in existing studies to collect traffic volume and vehicle movements data. Chakraborty et al. (2019) employed the speed radar gun to measure and collect vehicle speed data, which is used to estimate RTA occurrence frequency. Other than that, Basso et al. (2018) utilized the data collected through Automatic Vehicle Identification (AVI) systems installed at free-flow toll gates. The data is used to compute the traffic flow variables such as the flow of each vehicle type, vehicle speeds, changes in vehicle speed, variance of vehicle speeds, traffic density and changes in traffic density,

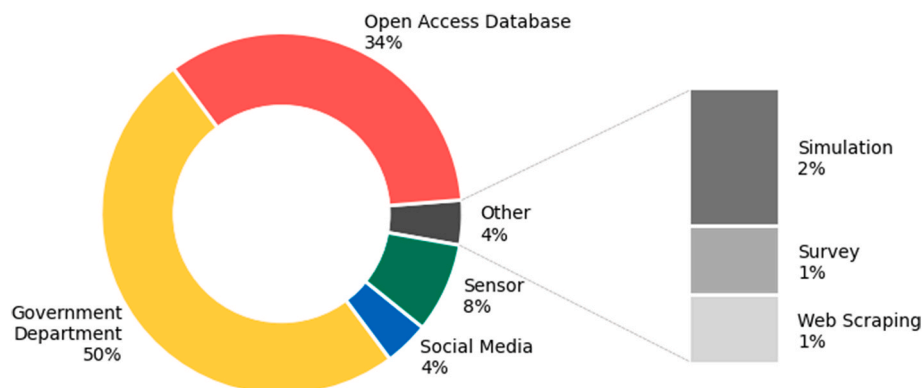


Fig. 3. Non-visual RTA data sources for ML-based RTA prediction.

to predict the occurrence of RTA. Researchers also sourced the data collected by sensors through existing tools such as the Car2X-Communication system that tracks emergency braking events (Reichenbach et al., 2021) and Signal Four (S4) analytics tool.

4.1.4. Social media

Social media is emerging as a new paradigm for RTA predictions with non-visual data, where individuals act as “sensors” that report traffic conditions on social media while traveling. For instance, some studies utilized the tweet contents from Twitter for various RTA tasks such as the RTA occurrence (Z. Zhang et al., 2018) and severity level (Raul et al., 2024) prediction based on unstructured social media data. However, the raw social media data requires a specific pre-processing process due to social media data, which can be noisy and unreliable for RTA predictions. The pre-processing is vital to filter the raw social media data so that only the RTA-related contents are included for RTA predictions. For instance, the Sensing Feature Extraction and View Construction (SFEVC) and Traffic Risk Feature Extraction (TRFE) are introduced by Zhang et al. (2019a, 2019b), respectively to filter the tweet contents and extract only the RTA-related features for the estimation of RTA risk level. Besides (Z. Zhang et al., 2018) include a filtration process to extract the tweet contents with RTA-related keywords such as “accident” and “crash”. These relevant tweets are then brought into the tokenization process for detecting RTA occurrence with the developed models.

4.1.5. Other

In addition to the previously discussed data sources, there are other sources being used occasionally in existing studies, including surveys, web scraping, and simulation data. The survey is a potential data source to obtain RTA-related data when there is a lack of information from the police reports. Specifically, Chakraborty et al. (2019) conducted a road inventory survey due to the lack of information in police recorded crash data prepared by Kolkata traffic police, which included only the time, geographic location, and severity of the historical RTAs. Through the road inventory survey, the researchers collected data describing road geometry, existing infrastructure, and traffic signal details to forecast the frequency of pedestrian-involved fatal RTA.

Apart from that, web scraping tools are also used to automate the extraction of RTA-related data from web resources, such as newspapers. For example, Siddik et al. (2021) utilized the Google extension “Web Scraper” tool to extract RTA-related content from Prothom Alo, a national newspaper in Bangladesh, due to the limited RTA datasets available. Using this technique, the researchers collected 1237 data samples of RTAs that occurred between 2013 and 2019 in Bangladesh. This data is used to train and validate the performance of the developed models in classifying the fatal and non-fatal RTA.

Under specific circumstances, it is difficult to collect real-world vehicle information, and thus, simulation data has been used as part of the non-visual RTA datasets to develop the RTA prediction models. Dogru and Subasi (2018) employed the Simulation of Urban Mobility (SUMO) to simulate and generate information on a vehicle’s velocity and position, due to the difficulty of collecting them in real-world situations. The simulation data from SUMO is incorporated into the study to predict the occurrence of RTAs using various ML algorithms, including SVM, ANN, and RF.

4.1.6. Critical analysis

The RTA data sourced from government departments and open-access databases are often the historical records describing the RTA that occurred in real situations. Therefore, they are believed to be beneficial for the models to learn and perform a more accurate prediction as compared to the models trained with simulated RTA data. However, a potential drawback of the historical RTA dataset is the issue of class imbalance. This issue is observed through the uneven distribution of the data instances among the severity levels. For example, the data instances under the category of non-injury usually come with a

larger size as compared to more severe categories such as serious injury and fatal. This introduces the potential for a model to overfit the majority and underfit the minority. Hence, data treatment techniques such as sampling or adjusting class weight are necessary to resolve the issue. Conversely, the class imbalance issue faced by the historical RTA datasets can be viewed as a strength of the simulated RTA data. RTA simulation serves as a convenient and replicable method to collect RTA data but is limited under a pre-defined simulation environment that might not cover all the possible real-world scenarios. This is observed through the COM performed with RTA data from the driving simulator experiment (Elamrani Abou Elasad et al., 2020). The simulation is conducted in a safe environment to experimentally collect RTA data with varying conditions such as weather, terrain, and traffic. However, the reliability of the collected data is questioned as it highly relies on the tasks assigned during the simulation. Considering the strengths and limitations of both historical and simulated RTA datasets, they are recommended sources to use in future research and the decision is highly dependent on the desired outcomes of the studies.

While the development of models with historical RTA records helps to analyze the characteristics and identify key factors of RTA, it is also vital to consider the feasibility of implementing the models in practice. To clarify, the historical RTA datasets are used to train CSM models but there is limited discussion on how the necessary RTA features can be collected as inputs for ad-hoc severity classification. Thus, it is essential to further explore the potential solution to realize it. In fact, the installation of sensors can be a solution due to their ability to perform instantaneous traffic data collection. With well-calibrated sensors, it is possible to obtain accurate traffic information such as the traffic flow, volume, and occupancy in real-time. Nevertheless, the feasibility of this solution for worldwide implementation can be a topic worth investigating, particularly for the regions that cannot afford the installation and maintenance cost to implement a complete sensory system. In the absence of an advanced traffic management system, it can be challenging to adopt solutions that prioritize the collection of real-time traffic data. As an alternative, some insights can be gained from the cases that used social media data for RTA prediction. The idea of utilizing humans as “sensors” could be an alternative for collecting near real-time RTA data. To summarize, alternative data collection methods should be considered in regions that lack advanced traffic management systems. For instance, a smartphone-based mobile application is more practical and efficient for the public to report RTAs in real-time as compared to social media.

The survey is a less common source being used to collect RTA data due to the level of difficulty in obtaining adequate samples. Basically, collecting RTA data through surveys is often associated with extensive interviews or questionnaires which might take more time to complete. In addition, the responses collected from the individuals are mostly their driving behaviors or experiences with RTA which can be subjective. This leads to the issue of biases as some individuals might hold back their mistakes or unsafe driving behaviors, resulting in a less accurate reflection on the crash scenarios. Thus, survey is not a recommended source to obtain high-quality RTA datasets, but it is an alternative for investigation at the regions where none of the other data sources are available.

4.2. Evaluation metrics

After discussing the RTA data sources, this section attempts to review the existing studies from the perspectives of the performance metrics for model evaluation. Discussion in Section 4 has grouped different RTA prediction tasks into the categories of CRM, COM, CSM, and CFM. Among these categories, the COM and CSM are related to classification tasks, while CRM and CFM are related to regression tasks in most cases. Different metrics can be used to measure the model performance based on the task’s characteristics. For instance, the main objective of classification tasks is to classify the RTA data instances into discrete

categorical outputs. Hence, the classification metrics are meant to measure the ability of the models to output a correct label or class category based on a set of attributes input to it. In contrast, the regression tasks attempt to perform a forecast of continuous numerical values. With this, the regression metrics measure the ability of the models to output an accurate estimation that fits the actual values. Typically, the regression metrics measure the extent of errors between the estimation and actual values.

4.2.1. Classification metrics

The performance metrics from the classification category are commonly found in the studies that developed models for discrete RTA classification. The classification metrics applied in the existing studies for performance evaluation include accuracy, precision, F1-score, recall (or sensitivity), area under the receiver operating characteristics (ROC) curve (AUC), specificity, false positive rate (FPR), geometric mean (G-Mean), average precision-recall score (APR score), Matthews' correlation coefficient (MCC), loss, negative predictive value (NPV), and positive predictive value (PPV). Specifically, the most notable metrics include accuracy (44 studies), precision (33 studies), F1-score (29 studies), recall (28 studies), and AUC (16 studies). These metrics are usually measured with the support of 4 elements, namely the true positive (TP), false positive (FP), true negative (TN), and false negative (FN).

- **Accuracy:** The accuracy of a model evaluates the model's ability to classify the inputs into their respective class categories correctly. It is often applied for CSM and COM where the accuracy is measured based on the ratio between the number of correctly classified data in terms of severity level or crash occurrence and the total number of instances.
- **Precision:** The precision of a model describes the quality, or level of relevancy when the input is classified as a TP. It is measured by considering the ratio between the TP and the total number of instances labeled as positive (both TP and FP).
- **Recall:** Recall, also known as sensitivity, is used to evaluate the model's ability to identify the TP instances. It can be calculated by taking the ratio of TP and all actual positives (TP and FN).
- **F1-score:** The F1-score is considered the harmonic mean for both precision and recall. This is a useful metric to measure the model's performance when the RTA prediction requires equal consideration of both precision and recall.
- **AUC:** The AUC is obtained by calculating the area under the ROC curve, which plots the true positive rate (TPR) against the FPR. It provides an overall indication of the model's ability to differentiate between positive and negative data instances for the different crash occurrence and severity level categories. A higher AUC is desirable because this informs that the model is better at categorizing the data into positive and negative instances.

While classification metrics are commonly used in studies that focus on RTA classification, there are cases where these metrics are used to evaluate the performance of the models performing regression tasks such as risk prediction. The model receives RTA features input to estimate the risk score, which ranges between 0 and 1 (Li et al., 2020; Zhao et al., 2019, 2020). Further processing is then performed to convert the output into categorical (low-risk and high-risk). This manipulation allows the use of classification metrics such as AUC to measure the model's ability to differentiate between low-risk and high-risk cases. In addition to AUC (Li et al., 2020) also incorporated the TPR and FPR to evaluate and benchmark the performance of the developed models in terms of their ability to produce TPs while avoiding FPs when categorizing the RTA risk scores.

4.2.2. Regression metrics

Regression metrics are commonly applied in studies that focus on

statistical and numerical forecasting such as CRM and CFM. The existing studies have leveraged various regression metrics such as mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), coefficient of determination (R-squared), overfitting ratio (OR), Pearson correlation coefficient (R), relative absolute error (RAE), root mean squared logarithmic error (RMSLE), root relative squared error (RRSE), and total root mean squared deviation (RMSD) to measure the performance of proposed models. The MAE, MSE, RMSE, and R-squared are the common selection for regression tasks.

- **MAE:** The MAE is used to measure the average absolute difference between the estimated and actual values. In CFM, this illustrates the average deviation on the predicted RTA frequency as compared to the actual RTA frequency.
- **MSE:** The MSE is obtained by taking the mean of the squared differences between the estimation and actual values. With the squaring of the errors, minimizing this metric is advantageous to reduce the probability of encountering extremely large errors in the model's prediction.
- **RMSE:** RMSE is derived by taking the squared root of MSE to measure the errors with the same magnitude as the original data (before squaring). This allows the evaluation of performance at the original scale while maintaining the ability to identify extremely large errors.
- **R-squared:** R-squared helps to identify the amount of variance on the output that can be predicted from the features input to the model. With a range between 0 and 1, a higher value of R-squared indicates that the model has a better fit to handle variance in the output.

While regression metrics are primarily designed for numerical prediction, there are exceptional cases where these metrics are adopted for classification tasks, such as CSM that defines the RTA severity level as regression problems (Haruna et al., 2020; Kumeda et al., 2019; Lee et al., 2020; Lin et al., 2020; Sarkar and Sarkar, 2020; C. Zhang et al., 2020). In these cases, the categories of RTA severity levels are converted from categorical to numerical, allowing the representation of the predicted RTA severity level in numerical form. This configuration enables performance to be measured based on the differences (errors) between the predicted outputs and the numerical values that represent the actual severity level. The common regression metrics used to assess the model's performance include MAE, MSE, and RMSE.

4.3. Machine learning approaches

Having explored the data sources and performance metrics used in RTA prediction, this section continues the discussion with the trend of ML algorithms investigated in recent studies, addressing Research Questions 2 and 3 as outlined in Section 3.1. With this, Table 3 is presented to show the comparison among all literature, in terms of datasets, prediction outputs, and ML approaches. It serves as a reference to understand and analyze the performance, strengths, and limitations of the ML algorithms. Methodologically, the algorithms can be grouped into 4 categories of ML approach, comprising statistical regression, conventional ML, deep learning, and hybrid approaches.

4.3.1. Statistical regression

Algorithms from the statistical regression category, such as the logistic regression (LGR) is an approach that uses mathematical equations to represent the relationships between the RTA features and their contributions towards the target variable. The primary goal of statistical regression algorithms is to obtain the coefficients of each RTA feature input and derive the equation of the best-fit line that predicts the target variable. In addition, it offers prediction with high interpretability where the coefficients are often used to identify the significant variables contributed to the outputs. For instance, numerous studies demonstrate the capability of LGR to perform informative prediction (Basso et al., 2018; Ghasedi et al., 2021; Infante et al., 2022; Rezapour et al., 2020).

Table 3

Comparison of the literature in terms of datasets, prediction outputs, and ML approaches.

Code	Dataset Size	Dataset Split Ratio (Train: Test:Validation)	Prediction Outputs	Best Model			Comparative Algorithms
				Algorithm	Category	Performance Metrics	
S01	358,962	80:20	Non-injury, Minor injury, Severe injury	RF	C	Accuracy (0.953), Sensitivity (0.948), Specificity (0.974), AUC (0.961)	XGBoost, DNN, CatBoost
S02	378,820		Fatal, Serious injury, Minor injury, Non-injury	MobileNet	D	Accuracy (0.982), Precision (0.983), Recall (0.989), F1-score (0.985)	MLP, CNN, LSTM, ResNet, EfficientNetB4, InceptionV3, Xception
S03		80:20	Accident frequency (Number of vehicles)	ARIMA	S	MAE (15.560)	SARIMAX
S04	408,800		Crash risk	VT-NET	D	14*8 grids: MSE (3.890), MAE (1.476), R2 (0.836) 28*16 grids: MSE (2.381), MAE (1.681), R2 (0.796)	ARIMA, LSTM
S05	10,458	80:20	Fatal, Injury, Property damage only	ResNet-50	D	Accuracy (0.835), Precision (0.857), Recall (0.862), Specificity (0.909), FPR (0.091), FNR (0.413), F1-score (0.860)	CART, KNN, SVM, XGBoost, GB, NB
S06	1,233,952	80:20	1 (Crash), 0 (Non-crash)	LSTM-CNN	H	AUC (0.963), TPR (0.937), FPR (0.116)	SSAE-LSTM, CatBoost
S07	2000		Fatal, Major, Minor, No severity	WESTDT	C	Accuracy (0.854), Precision (0.752), Recall (0.778)	DT, RESSEL, STDT
S08	1916	80:20	Minor, Moderate, Major, Catastrophic	XGBoost	C	Accuracy (0.790), Precision (0.806), Recall (0.792), F1-score (0.795)	LightGBM, CatBoost, RF, SVM, MNL
S09	170,343	70:30	Injury, No injury, Possible injury	XGBoost, CatBoost	C	XGBoost: accuracy (0.760), precision (0.687), recall (0.757) CatBoost: accuracy (0.760), precision (0.688), recall (0.757)	RF, LightGBM
S10	17,495		1 (Crash expected in next hour), 0 (Otherwise)	DSHN	H	Accuracy (0.757), Recall (0.722), AUC (0.798), FAR (0.215)	ANN, CNN, LSTM, Bi-LSTM, LSTM-CNN, STCL-Net
S11		70:30	Severe, Non-severe	MLP	D	Accuracy (0.735), Precision (0.763), Recall (0.706), F1-Score (0.733)	TabNet
S12	70,962		Fatal, Serious injury, Slight injury	ORF	C	G-Mean (0.478)	RF, ONF, OL, XGBoost
S13	16,122		Injury, Fatal	SVM	C	Accuracy (0.960), Sensitivity (0.930), Specificity (1.000), G-Mean (0.960)	KNN, DT
S14			Accident severity (No category details given)	RF	C	Accuracy (0.856), Precision (0.920), Recall (0.870), F1-Score (0.890), AUC (0.880)	NB, LGR, KNN, SVM
S15	5,860,405		Fatal, Non-fatal	CNN	D	Accuracy (0.755), Precision (0.757), Recall (0.755), F1-Score (0.756)	RF, KNN, SVM
S16	28,103	67:33	Property damage only, With victims (Minor injury, serious injury or death)	LGR	S	Accuracy (0.652), F1-Score (0.277), Sensitivity (0.732), Specificity (0.645), PPV (0.171), NPV (0.960), G-Mean (0.687), MCC (0.222)	RF, NB, KNN, DT, SVM
S17	4946	2:1	Crash, Non-crash	CNN	D	Accuracy (0.795), Precision (0.800), Recall (0.790), F1-Score (0.795)	BLR
S18	1559	70:30	Injury, Death	ANN	D	AUC (0.753)	ANN
S19	65	85:15	Number of fatal accidents	RF	C	R-squared (0.945), MAPE (2.600)	SARIMA, SARIMAX
S20		75:25	Injured, Killed	GBM	C	Accuracy (0.890), Precision (0.890), Recall (0.890), F1-Score (0.885), AUC (0.950)	XGBoost, RF, SVM
S21	68,665	50:50	Crash, Near crash, Normal driving	CNN-GRU-XGBoost	H	Accuracy (0.975), Precision (0.847), Recall (0.713), F1-Score (0.774), AUC (0.974), FPR (0.008)	XGBoost, MLP, RF, LGR, AdaBoost
S22	52,843		No pedestrian death, Pedestrian death	BNN	D	Accuracy (0.894), F1-Score (0.940), AUC (0.840), MCC (0.540)	RF, NN, BN
S23	51,871	70:30	No damage, Minor damage, Functional damage, Disabling damage	XGBoost	C	Accuracy (0.924), Loss (0.180)	MLR, XGBoost, ET, RF, NB
S24	2165		Crash risk level, Crash occurrence (Yes/No)	RLT	C	CRM: AUC (0.903), TPR (0.793), FPR (0.100) COM: AUC (0.936), TPR (0.927), FPR (0.100)	LGR, SVM, DNN
S25	78,227	70:30	Slight traffic delay, Serious traffic delay	LightGBM	C	F1-Score (0.678), AUC (0.884), APR Score (0.731), G-Mean (0.776)	RF
S26	41,953	80:20	Slight (None and slight injury), Serious (Fatal and serious injury)	RF	C	Accuracy (0.830), F1-Score (0.820), AUC (0.910)	NB, LGR, KNN, DT
S27		70:15:15	Number of accidents	ANN	D	R-squared (0.999)	MVR
S28	2203	70:30	Fatal, Serious, Slight	PSO-ANN	H	Accuracy (0.853), Precision (0.820), Recall (0.810), F1-Score (0.830)	RF, NB, KNN, NC
S29	12,316	80:20	Fatal, Serious, Slight	RF	C	Accuracy (0.940), Precision (0.941), Recall (0.933), F1-Score (0.938)	NB, LGR, KNN, DT, SVM, AdaBoost
S30	22,576	70:30	Slight injury, Serious injury	RF	C	Accuracy (0.973), Precision (0.980), F1-Score (0.970), AUC (0.970), Sensitivity	KNN, DT, AdaBoost

(continued on next page)

Table 3 (continued)

Code	Dataset Size	Dataset Split Ratio (Train: Test:Validation)	Prediction Outputs	Best Model			Comparative Algorithms
				Algorithm	Category	Performance Metrics	
S31	57,472	80:20	No indication of injury, Non-incapacitating injury, Incapacitating injury	CNN	D	(0.970), Specificity (0.970), G-Mean (0.970) Accuracy (0.860), Recall (0.857), Sensitivity (0.764)	LGR, DT, SVM
S32	1061	70:15:15, 80:10:10	Vehicle accidents: Property damage only, Injury/Fatality Pedestrian accidents: Injury, Fatal	ANN	S	Vehicle accidents: Accuracy (0.829) Pedestrian accidents: Accuracy (0.924)	LGR
S33	2743		No injury, Minor injury, Severe injury, Fatal injury	DJ	C	Accuracy (0.837), Precision (0.850), Recall (0.730)	RF, DT
S34	10,048	80:20	Fatal, Injury, Property damage only	CNN	D	Precision (0.790), Recall (0.840), F1-Score (0.760)	SVM
S35	110,000	70:15:15	Traffic incident, No traffic incident	RF	C	Accuracy (0.848), Precision (0.447), Recall (0.685), AUC (0.870)	NN, LGR, DT
S36	448,511	70:30	1 (Shortest traffic delay), 2, 3, 4 (Longest traffic delay)	RF	C	Accuracy (0.952), Precision (0.593), Recall (0.623), F1-Score (0.603)	MLR, XGBoost, DT
S37	1237	70:30	Death, No death	DT	C	Accuracy (0.850), Precision (0.880), Recall (0.850), F1-Score (0.850)	NB, LGR, KNN
S38	5479 (15 years)	13:1:1	Number of accidents	Bi-LSTM-Attention	H	MSE (4.000), MAE (0.910), RMSE (2.000)	Prophet, DNN, LSTM, Bi-LSTM
S39	140,690		Class 1 (Non-injury), Class 2 (Complaint of pain, Other visible injury), Class 3 (Severe injury, Killed)	XGBoost	C	Precision (0.680), Recall (0.643), F1-Score (0.647)	MLP, SVM
S40	4070	70:30	Property damage only, Complaint of pain, Visible injury, Severe injury, Fatal	ANN	D	Accuracy (0.616)	GA-ANN
S41	37,774	70:30	Serious injuries or fatal, Slight	MLP	D	Accuracy (0.827), Precision (0.841), F1-Score (0.808), Sensitivity (0.795)	SVM
S42	10,000	70:30	Severe, Non-severe	SVM-FCM	H	Accuracy (0.742)	FNN, FNN-FCM, SVM
S43	54,568	70:15:15	Non-injury, Injury, Incapacitating injury	PSO-ANN	H	Accuracy (0.736), MSE (0.196)	NN
S44	6,750,072	70:30	Crash, Non-crash	CNN	D	AUC (0.956), Specificity (0.907), Sensitivity (0.888)	LGR, SVM, ANN
S45		90:10	No or minor injury, Heavy wound or death	RF	C	Accuracy (0.732), APR Score (0.670)	SGD Classifier, XGBoost, MLP, NB, LGR, KNN, DT
S46			Crash, Non-crash	MLP	D	Accuracy (0.933), Precision (0.873), Recall (0.938), G-Mean (0.934), F1-Score (0.903), AUC (0.932)	SVM
S47	6515	70:30	Fatal or injury, Property damage only	LGR	S	Precision (0.792), Recall (0.625), F1-Score (0.681), TPR (0.625), FPR (0.458), TNR (0.542)	RF, KNN, DT
S48	A: 7,736, B: 7424		Emergency call relevant (Possible, serious, or fatal injury), Emergency call not relevant (Non-injury)	SVM	H	Accuracy (0.925), Precision (0.993), Recall (0.847), F1-Score (0.914)	RF, LGR, DNN
S49	6565		Major injury, Minor injury, Fatality	DPSO-FCBF-FLANN BP	H	Accuracy (0.897), MSE (0.051)	FLANN BP
S50	894,086	80:20	Crash, Non-crash	CNN	D	Accuracy (0.773), F1-Score (0.765), AUC (0.846), APR Score (0.870)	RF, LGR, KNN, DT, SVM
S51	4299	80:10:10	Crash, Non-crash	LSTM	D	Accuracy (0.788), FAR (0.171)	KNN, LGR, NB, ANN, SVM, RF
S52	959		Fatal, Injury, No injury	DT-RFL	H	Accuracy (0.700), Precision (0.675), Recall (0.630), F1-Score (0.720)	DT, DT-CFL
S53	518	75:25	Fatal, Serious injury, Minor injury	RF	C	MSE (0.047), RMSE (0.217)	ANN, DT
S54	525,600	75:25	Accident risk (Probability)	LSTM-CNN	H	AUC (0.930), FPR (0.120), Sensitivity (0.880)	XGBoost, CNN, LGR, LSTM
S55	8859	75:25	Not injured, Non-incapacitating injury, Possible injury, Suspected serious injury, Killed	XGBoost	C	MAE (0.714)	RF
S56	7654	75:25	Non-injury, Simple injury. Grievous injury, Fatal injury	CIF	C	Precision (0.470), Recall (0.440), F1-Score (0.450), Test error (46.000)	OP, RF
S57	1919		Property damage only, Injury/Fatality	BLR	S	Accuracy (0.690), Precision (0.790), Recall (0.730), R-squared (0.600)	CT
S58	133	70:30	Fatal, Non-fatal	ANN	D	Accuracy (0.910), Precision (0.893), MSE (0.156), Sensitivity (0.962), Specificity (0.836)	MLR
S59	222	67:33	Number of accidents	DNN	D	MAE (3.927), RMSE (5.908), R-squared (Coefficient of determination) (0.945)	RENB, GEP

(continued on next page)

Table 3 (continued)

Code	Dataset Size	Dataset Split Ratio (Train: Test:Validation)	Prediction Outputs	Best Model			Comparative Algorithms
				Algorithm	Category	Performance Metrics	
S60	156	70:30	Minor injury, Serious injury, Severe injury	RF	C	Accuracy (0.948), Precision (0.949), Recall (0.948), F1-Score (0.948), TPR (0.948), FPR (0.025)	MLP, NB, DT, SVM, MNL
S61	8516		Fatal, Hospitalized, Injured, Property damage only	DT	C	Accuracy (0.738), Precision (0.759), Recall (0.738), AUC (0.895), TPR (0.738), FPR (0.159)	MLP, PART
S62		70:30	Number of accidents	SVR	C	MSE (0.029), R-squared (0.9874)	NN
S63	5000	70:30	Fatal, Severe injury, Light injury	K-Means-RF	H	Accuracy (0.999), Precision (1.000), Recall (1.000), F1-Score (1.000)	LGR, KNN, SVM, ANN
S64			Number of casualties	LSTM-GBRT	H	RMSLE (0.017), R-squared (0.996)	NN, MNL, GBRT, LSTM
S65			Accident risk (Probability)	CNN-RF	H	AUC (0.992)	CNN
S66	271,563	70:30	Fatal or serious injury, Minor injury	RF	C	Accuracy (0.755), Precision (0.881), Recall (0.591), F1-Score (0.707), AUC (0.755)	NB, LGR, AdaBoost
S67			Number of fatal pedestrian crashes	BRNN	D	MAE (0.312), RMSE (0.402), R (0.585), R-squared (0.342), OR (1.005)	LMNN, SCGNN, GDXNN
S68	555	90:10	Fatal, Serious injury, Slight injury	FARC-HD	H	Accuracy (0.859), Precision (0.739), Recall (0.859), F1-Score (0.795), TPR (0.859), FPR (0.859), MAE (0.160), RMSE (0.286), RAE (98.980), RRSE (99.990)	RF, HLVQ, RBF, MLP, NB
S69	43,089	70:30	Fatal, Grievous injury, Simple injury, Motor collision	Adaboost	C	Accuracy (0.800), Precision (0.750), F1-Score (0.740), G-Mean (0.930)	NB, KNN, DT
S70	655,643	80:20, 90:10	Injury (Fatal or Injury), Property damage only	HS-SVM	H	Accuracy (0.835), Precision (0.984), AUC (0.845), Sensitivity (0.860), Specificity (0.833)	WOA-SVM, SVM, PSO-SVM, MXL
S71	1525		No injury, Slight, Serious, Killed	XGBoost	C	Accuracy (0.831), Precision (0.828), Recall (0.827), F1-Score (0.824)	MLR
S72	85,214	65:35	Accident, Non-accident	PNN	D	Accuracy (0.990), TPR (0.800), FPR (0.005)	SVM
S73	876	75:25	Crash, Non-crash	DNN	D	Accuracy (0.690), AUC (0.641), TPR (0.521), TNR (0.770)	RF, NB, KNN, DT, SVM, SNN
S74	8516		Property damage, Injured, Hospitalized, Fatal	RF	C	Precision (0.763), Recall (0.624), AUC (0.908), TPR (0.624), FPR (0.104)	KNN, DT, MNL
S75	201,581	80:20	Property damage only, Injury, Fatality	RF	C	Precision (0.660), Recall (0.790), F1-Score (0.720)	MNL
S76	4320	30:70	Property damage only, Injury, Fatal	RST-SVM	H	Accuracy (0.944)	SVM
S77	8535		Fatal injury, Non-fatal injury	NB	C	AUC (0.709)	KNN
S78			Accident risk (Rate)	RiskCasts	H	MAE (0.125)	MLP, LIR, RR
S79	239,734		Traffic risk index	DeepRisk	H	MAE (0.949)	MLP, LIR, RAND, ARD, LASSO
S80	797,462	80:20	Accident risk (Probability)	FMCNN	H	AUC (0.936)	FM, DNN
S81	21,436	80:20	Slight, Serious, Fatal	CNN	D	Precision (0.893), Recall (0.932), F1-Score (0.840)	NB, NN, LGR, KNN, DT, SVM, LSTM
S82	49,751	66:34	Fatal, Serious, Slight	RF	C	Accuracy (0.807), Precision (0.814), Recall (0.806), F1-Score (0.801)	ANN, DT, SVM
S83	3050		Fatal, Serious, Slight	NB	C	Accuracy (0.946), AUC (0.900)	NB
S84	13,029	80:20	Accident, Non-accident	LGR	S	FPR (0.209), Sensitivity (0.679)	SVM
S85	4279	90:10	Accident, Non-accident	RF	C	Accuracy (0.920), Sensitivity (0.940), Specificity (0.880)	ANN, SVM
S86	5 years data	4 years:1 year	Number of accidents	DAE-MVNB	H	MAE (0.150), Total RMSD (43.652%)	SVM
S87			Number of traffic deaths	PSOM-SVM	H	MSE (0.036), R-squared (0.973)	NN, KNN, SVM, PSOM-BPNN, PSO-SVM, BN
S88	13,673	70:30	Fatal, Serious injury, Slight injury	GA	C	Accuracy (0.870), Precision (0.860), Recall (0.880)	ANN, KNN, DT, SVM, BN
S89	5538	80:20	No injury. Possible/invisible injury, Non-capacitating injury, Incapacitating injury, Fatal injury	RF	C	Accuracy (0.539)	KNN, DT, SVM, OP, MNL
S90	297,113	70:30	Fatal injury, Incapacitating injury, Non-incapacitating injury, Possible injury, Non injury	DT	C	Accuracy (0.847), G-Mean (0.080)	GBM, RF, NB, NN, LGR
S91	281,716		High crash rate (H), Low crash rate (R)	CART	C	Accuracy (0.700), Precision (0.740), Recall (0.699), F1-Score (0.695)	ANN, SVM
S92	16,548		Crash, Non-crash	GBDT	C	Precision (0.011), Recall (0.764)	XGBoost, RF
S93	2,222,548	72:10:18	Accident risk (Frequency)	LSTM	D	MSE (0.001), MAE (0.014), RMSE (0.034)	RF, MLP, SVR, DT, RR, LASSO, ARMA
S94	222	67:33	Number of accidents	SVM	C	MAE (3.788), RMSE (5.905), R-squared (0.945)	RENB
S95	2700		Accident, Non-accident	DBN	D	Accuracy (0.850), Precision (0.930), Recall (0.710)	ANN, SVM, sLDA, LSTM

Best Model-Algorithm, Comparative Algorithms: AdaBoost = Adaptive Boosting; ANN = Artificial Neural Network; ARD = Automatic Relevance Determination; ARIMA = Autoregressive Integrated Moving Average; ARMA = Autoregressive Moving Average; Bi-LSTM = Bidirectional Long Short-Term Memory; Bi-LSTM-Attention = Bidirectional Long Short-Term Memory with Attention Layer; BLR = Binary Logistic Regression; BN = Bayesian Network; BNN = Bayesian Neural Network; BRNN = Bayesian Regularization Neural Network; CART = Classification and Regression Trees; CatBoost = Categorical Boosting; CIF = Conditional Inference Forest; CRPOP = Correlated Random Parameters Ordered Probit; CT = Classification Tree; CNN = Convolutional Neural Network; CNN-GRU-XGBoost = hybrid of Convolutional Neural Network, Gated Recurrent Unit and Extreme Gradient Boosting; CNN-RF = hybrid of Convolutional Neural Network and Random Forest; DAE-MVNB = Hybrid of Denoising Autoencoder and Multivariate Negative Binomial; DBN = Deep Belief Network; DeepRisk = A deep transfer learning scheme for traffic risk estimation with social sensing data; DJ = Decision Jungle; DNN = Deep Neural Network; DPSO-FCBF FLANN BP = Hybrid of Discrete Particle Swarm Optimization, Fast Correlation-Based Feature Selection and Functional Link Artificial Neural Network Back Propagation; DSHN = Deep Spatiotemporal Hybrid Network; DT = Decision Tree; DT-CFL = Decision Tree-based Converted Fuzzy Logic; DT-RFL = Decision Tree-based Revised Fuzzy Logic; ET = Extra Trees; FARC-HD = Fuzzy Association Rule-Based Classification Model for High-Dimensional Problems; FLANN BP = Functional Link Artificial Neural Network Back Propagation; FM = Factorization Machine; FMCNN = Hybrid of Factorization Machine and Convolutional Neural Network; FNN = Feedforward Neural Network; FNN-FCM = Fuzzy C-Means based Feedforward Neural Network; GA = Genetic Algorithm; GA-ANN = Genetic Algorithm-Artificial Neural Network Hybrid; GBDT = Gradient Boosted Decision Trees; GBM = Gradient Boosting Machine; GBRT = Gradient Boosted Regression Trees; GDXNN = Gradient Descent and Adaptive Learning Neural Network; GEP = Gene Expression Programming; HLVQ = Hierarchical Learning Vector Quantization; HS-SVM = Harmony Search-Support Vector Machine hybrid; RiskCast = Inductive multi-view learning approach for forecasting RTA risks; K-Means-RF = K-Means-Random Forest hybrid; KNN = K-Nearest Neighbors; LASSO = Least Absolute Shrinkage and Selection Operator Regression; LightGBM = Light Gradient Boosting Machine; LMNN = Levenberg-Marquardt Neural Network; LIR = Linear Regression; LGR = Logistic Regression; LSTM = Long Short Term Memory; LSTM-CNN = Long Short Term Memory-Convolutional Neural Network hybrid; LSTM-GBRT = Long Short Term Memory-Gradient Boosted Regression Trees hybrid; ML = Mixed Logit; MLP = Multilayer Perceptron; MLR = Multivariate Logistic Regression; MNL = Multinomial Logit; MUL = Multivariate Nonlinear Regression; MVR = Multivariate Regression; NB = Naïve Bayes; NC = Nearest Centroids; NN = Neural Network; OL = Ordered Logit; ONF = Ordered Non-honest Forest; OP = Ordered Probit; ORF = Ordered Forest; PART = Partial Decision Trees; PNN = Probabilistic Neural Network; PSO-ANN = Particle Swarm Optimization-Artificial Neural Network hybrid; PSOM-BPNN = Particle Swarm Optimization with Mutation-Back Propagation Neural Network hybrid; PSOM-SVM = Particle Swarm Optimization with Mutation-Support Vector Machine hybrid; PSO-SVM = Particle Swarm Optimization-Support Vector Machine hybrid; RAND = Random Based Estimation; RBF = Radial Basis Function Network; RENB = Random-Effect Negative Binomial; RESSEL = Reliable Semi-Supervised Ensemble Learning; RF = Random Forest; RLT = Reinforcement Learning Trees; RPOP = Random Parameters Ordered Probit; RR = Ridge Regression; RST-SVM = Rough Set Theory-Support Vector Machine hybrid; SARIMA = Seasonal Autoregressive Integrated Moving Average; SARIMAX = Multivariate SARIMA; SCGNN = Scaled Conjugate Gradient Learning Neural Network; SGD Classifier = Stochastic Gradient Descent classifier; sLDA = Supervised Latent Dirichlet Allocation; SNN = Shallow Neural Network; SSAFE-LSTM = Stacked Sparse Autoencoder-Long Short Term Memory; STCL-Net = Spatiotemporal Convolutional Long Short Term Memory Network; STDT = Self-Training with Decision Tree; SVM = Support Vector Machine; SVM FCM = Fuzzy C-Means based Support Vector Machine; SVR = Support Vector Regression; TabNet = An interpretable deep learning architecture for tabular data; WOA-SVM = Whale Optimization Algorithm-Support Vector Machine hybrid; XGBoost = Extreme Gradient Boosting.

Best Model-Category: S = Statistical Regression; C = Conventional ML; D = Deep Learning; H = Hybrid.

Best Model-Performance Metrics: AUC = Area under the receiver operating characteristics curve; APR Score = Average precision-recall score; FAR = False alarm rate; FPR = False positive rate; G-Mean = Geometric mean; MAE = Mean absolute error; MAPE = Mean absolute percentage error; MCC = Matthews correlation coefficient; MSE = Mean squared error; NPV = Negative predictive value; OR = Overfitting ratio; PPV = Positive predictive value; R = Pearson's correlation coefficient; R-squared = Coefficient of determination; RAE = Relative absolute error; RMSE = Root mean squared error; RMSLE = Root mean squared logarithmic error; RRSE = Relative root squared error; TNR = True negative rate; Total RMSD = Total root mean squared deviation; TPR = True positive rate.

The coefficients that form the modeling equation are useful for analyzing the significant risk factors contributing to RTAs. However, statistical regression is less commonly used in recent studies, where only sporadic cases have concluded that it outperforms other ML approaches. The statistical regression algorithm is well known for its interpretability and simplicity, but this becomes an issue in the context of RTA prediction. It mostly captures only the linear relationships but in fact, the RTA features exhibit complex relationships. Fiorentini and Losa (2020) developed an LGR-based model to classify the severity level based on the RTA features input to it. Although it achieves the best overall accuracy among the other baseline models, it has the worst performance in predicting specific crashes such as fatal and injury-related crashes which are associated with complex interactions between the RTA features. Furthermore, statistical regression algorithms also do not achieve satisfactory performance for regression tasks such as CRM and CFM which involve time series data. The temporal dependencies of time series data further magnify the limitation of statistical regression algorithms because they are unable to capture the dynamic temporal relationships of RTA features.

4.3.2. Conventional machine learning

Moving beyond statistical regression, conventional ML is the most widely investigated approach in the context of RTA prediction. In contrast to statistical regression, conventional ML is claimed to be better at learning complex patterns and adapting to the non-linear relationships between the RTA features. Common conventional ML algorithm used for regression includes tree-based algorithms (Liu et al., 2018; Yang et al., 2022), RF (Nassiri et al., 2022), and SVM (Singh et al., 2018; Wu and Wang, 2020), which outperforms the LGR, seasonal autoregressive integrated moving average (SARIMA), and random-effect negative binomial (RENB) from the statistical regression category. With a variety

of algorithms existing in the conventional ML category, there is no conclusive case to highlight the trend of algorithm selection because it requires thoughtful consideration of the characteristics of RTA datasets from different regions. However, literature shows that distinct focuses are put on the ensemble and gradient boosting algorithms for RTA prediction. Meanwhile, there are sporadic studies with other algorithms such as Naïve Bayes (NB) and genetic algorithm (GA).

Based on the literature, CSM is the most investigated task with conventional ML algorithms. Among the algorithms, RF is the most common algorithm concluded with satisfactory performance based on the literature conducted with RTA datasets from various regions such as the US, the UK, Ghana, Russia, Ethiopia, India, Germany, Senegal, Korea, and Yemen. RF is an ensemble ML technique that combines the predictive power of multiple weak learners, usually a series of DTs for boosted performance. This key innovation allows RF to handle various aspects of the non-linear relationships in RTA datasets and minimize overfitting, resulting in a more robust RTA prediction. To shed light on this, Tariq et al. (2020) compare the performance of statistical regression (MNL), conventional ML (NB, DT, SVM, and RF) and deep learning (MLP) in classifying the severity level of RTA in Yemen. All the conventional ML algorithms have better performance than the statistical regression and RF is the most accurate model in terms of FPR, TPR, precision, and recall. Nevertheless, the size of RTA dataset used in this study is small (156 instances). Hence, comparing RF with deep learning might not be as relevant because deep learning is primarily designed to excel in learning the complex relationships among features, particularly in large datasets.

In addition to ensemble, the gradient boosting technique has also shown promising performance in predicting RTA, particularly for tasks such as CSM and COM. The gradient boosting technique sequentially builds the models, and each model attempts to minimize the prediction

errors of their predecessor. This sequential learning empowers the gradient boosting algorithms to adapt to complex RTA relationships and improve prediction accuracy. Some of the gradient boosting algorithms include extreme gradient boosting (XGBoost), categorical boosting (CatBoost), adaptive boosting (AdaBoost), gradient boosting machine (GBM), light gradient boosting machine (LightGBM), and (gradient boosted decision trees) GBDT. Particularly, XGBoost is widely applied for RTA prediction. [Vadhwani and Thakor \(2022\)](#) utilized XGBoost algorithm to classify RTA severity level and concluded that it has the best performance with an accuracy of 0.924 as compared to multivariate logistic regression (MLR), extra trees (ET), RF, and NB. However, this is highly dependent on the quality of RTA datasets ([Meng et al., 2018](#)). [Shi et al., 2022; Zhao et al., 2020](#) showed that the GBDT has a better performance than XGBoost when the RTA dataset is imbalanced, with only 37 positive instances in 16,688 data samples. Without class imbalance treatment, the GBDT algorithm has proven its ability to perform prediction based on skewed variables with limited transformations.

4.3.3. Deep learning

Deep learning, categorized as a subset of ML involves the application of ANN to extract complex patterns and correlations from the input features. Recently, various deep learning algorithms have existed with unique architectures and strengths, providing flexibility to develop models for different RTA prediction tasks. This characteristic is beneficial to suit a variety of prediction tasks, enabling deep learning to outperform statistical regression. For instance, the Ridge Regression, Lasso Regression, Autoregressive Moving Average (ARMA), and RENB are inferior as compared to LSTM ([Ren et al., 2018](#)), DNN ([Singh et al., 2020](#)), and Bayesian Regularization Neural Network (BRNN) ([Chakraborty et al., 2019](#)) due to their limited ability to capture complex and non-linear relationships among the RTA features.

CNN and LSTM are two prominent deep learning algorithms that specialize in distinct types of data processing. Specifically, CNN is designed for the image-based application but has proven its applicability in detecting RTA occurrence and classifying severity levels using non-visual data ([Huang et al., 2020; Khan et al., 2024; Rahim and Hassan, 2021](#)). This is achievable through the numeric-to-image transformation technique to represent the non-visual RTA features in terms of images, a format that is well-suited for prediction with CNN. Conversely, LSTM is renowned for its ability to capture time series dependencies among RTA features, enabling the extraction of RTA patterns over specific time intervals. [Ren et al. \(2018\)](#) exemplified this by developing an LSTM-based model to estimate RTA risk within the predefined period. The results verified the superiority of LSTM in capturing temporal and regional spatial patterns from the RTA datasets. Furthermore, [Jiang et al. \(2020\)](#) developed a model to detect RTA occurrence using three LSTM networks, each designed to focus on different temporal resolutions (long, middle, and short). The improved performance of this model highlights the significance of considering various temporal resolutions to extract dynamic variations in traffic conditions. In summary, these successful cases demonstrate that deep learning algorithms, particularly CNN and LSTM have better capabilities in handling complex correlations, especially in the context of temporal and spatial perspectives which are crucial for RTA prediction.

4.3.4. Hybrid

The hybrid approach is an approach to developing RTA prediction models through the integration of multiple algorithms. Existing studies show that hybrid models are efficient in performing accurate RTA prediction, and this success encourages further exploration and research in the future. To clarify, better performance is achieved by leveraging the strengths of individual algorithms to complement the weaknesses of one another. Frequently, CNN, LSTM, and Particle Swarm Optimization (PSO) are integrated into the RTA prediction models. The CNN algorithm is included as part of the models due to its ability to extract meaningful features from input RTA data. One significant challenge to

working with tabular RTA data is the sparsity of the dataset which increases the difficulty to extract recognizable patterns or relationships among the features. Thus, CNN is adopted as a feature extractor, where the extracted features are then used as inputs for other ML algorithms such as RF and XGBoost to predict RTA occurrence or risk level ([Shi et al., 2022; Zhao et al., 2020](#)). Additionally, CNN is also employed to extract high-order ([Zhao et al., 2019](#)) and time-invariant ([Li et al., 2020](#)) feature interactions and combined with the extraction from other algorithms to estimate RTA risk levels. Specifically, the LSTM-CNN is an effective hybrid model that combines both long-term dependency and time-invariant features for boosted performance ([Li et al., 2020; Mo et al., 2024](#)).

While conventional ML and deep learning approaches have satisfactory performance, careful selection of the most optimal hyperparameters remains a challenging task to prevent underfitting or overfitting. Thus, optimization algorithms are involved in the hybrid approach to assist in hyperparameter tuning. For example, the PSO, with its variations such as Discrete PSO (DPSO) and PSO with Mutation (PSOM) are commonly integrated as part of the models for optimizing the hyperparameters of the RTA prediction model. The PSO algorithm is found to be supportive in obtaining the most optimum weight values of the neural networks-based RTA prediction models ([Ashikuzzaman et al., 2021; C. Zhang et al., 2020](#)). Moreover, this approach is effective for both the conventional ML and deep learning approaches. The integration of DPSO into the feature selection helped to reduce the overall processing time of the ANN model by approximately 2.12% ([Haruna et al., 2020](#)). In addition, the SVM model tuned with PSOM has experienced a reduction of MSE from 6.43% to 3.63%, indicating an improved performance with the hybrid model.

4.3.5. Critical analysis

Based on the analysis on existing literature, it is evident that innovative attempts have been made to discover the applicability of different ML approaches to enhance road safety. As technology evolves, various ML approaches are introduced to develop better solutions to enhance road safety. It started with statistical regression, followed by conventional ML, and progressed towards deep learning and hybrid approaches. Statistical regression models are proposed to replace the traditional statistical RTA analysis using graphs and charts. These models enabled various RTA tasks to be predicted using regression while providing foundational insights into the correlation between RTA features. However, they are inefficient in working with complex and high-dimensional RTA data. To address this, conventional ML techniques such as DT, RF and gradient boosting algorithms were introduced to capture non-linear relationships for more accurate RTA predictions. Technology advancements have led to the development of more innovative road safety solutions, where various deep learning algorithms are introduced to serve different purposes. Frequently used deep learning algorithms include CNN and LSTM, which respectively specialize in extracting spatial and temporal features. With more mature algorithms, recent literature has explored novel combinations of multiple algorithms as hybrid models to enhance the RTA prediction tasks. The hybrid approach aims to integrate the strengths of these algorithms to achieve highly accurate RTA predictions for improved road safety.

Despite the innovative aspects of these ML approaches, it is vital to summarize their advantages and limitations to suit the application in the RTA prediction context. For this, [Fig. 4](#) compared the frequency that the ML approaches are investigated and concluded as the best model for RTA predictions. With close reference to the figure, some statistical insights can be obtained to determine the emerging opportunities for future research. As discussed, statistical regression is valued for its simplicity and high interpretability but is less for datasets that exhibit non-linear and complex relationships. It was investigated in 49 recent literature and only 5 studies (10%) concluded it as the best suit model for RTA prediction tasks. For the conventional ML approaches, they have gained significant attention and consensus in recent literature for their

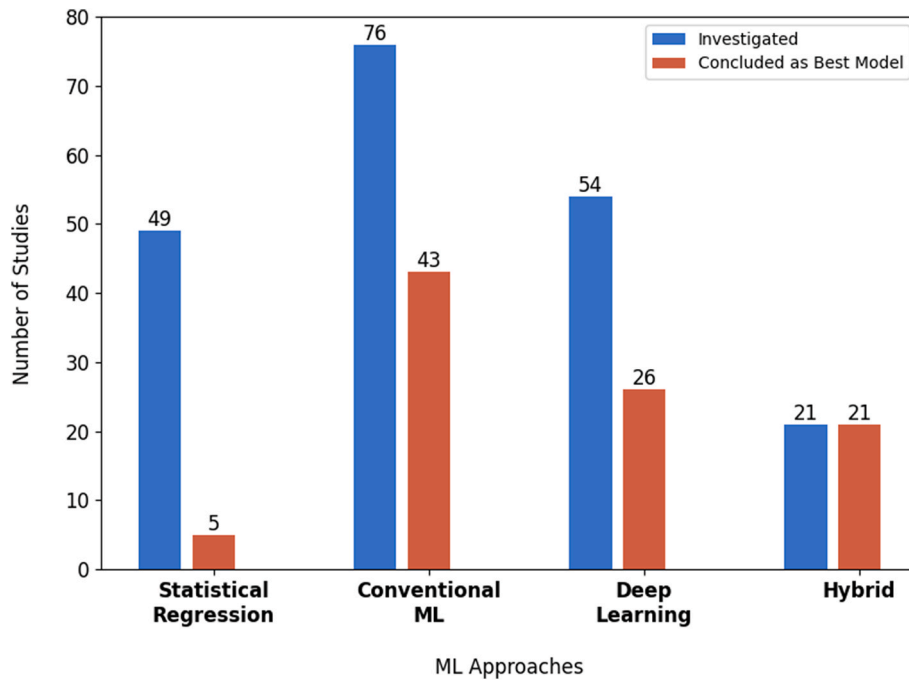


Fig. 4. ML approach selection trends: Frequency of investigation and concluded as best model.

superior performance in predicting RTAs. Statistics showed that conventional ML models have outperformed the others in 43 out of 76 studies (57%). It is the most widely investigated category and can be seen as a more mature approach to handle complex RTA features effectively.

The emerging opportunity for further exploration can focus on advanced methods such as deep learning and hybrid models. The deep learning and hybrid approaches are emerging paradigms that received less investigation compared to the conventional ML approach. Undeniably, more research is recommended to evaluate their effectiveness to work with larger RTA datasets and more complex RTA relationships. Moreover, it is noteworthy that deep learning often provides limited prediction interpretability, which is crucial for analyzing significant RTA factors. There are specific deep learning algorithms such as TabNet that can provide interpretable predictions. The TabNet-based RTA prediction model can achieve comparable performance while offering interpretability (Sattar et al., 2023), but its long training time requires the integration of other algorithms. The hybrid approach is potentially useful by adopting algorithms for tasks such as feature selection and hyperparameter tuning to reduce the training time and development efforts. Furthermore, it is imperative to further investigate the major challenges faced by the hybrid approach such as additional complexity and computational power requirements. In conclusion, the suitability of different ML approaches depends on the RTA prediction tasks. Hence, Table 4 is presented to outline the strengths and limitations of each ML approach, followed by their suggested use cases.

4.4. Challenges and future directions

Analysis from previous discussions has outlined the existing gaps and ignited potential research directions. Therefore, this section intends to summarize the challenges of RTA prediction with non-visual data and propose future research directions, aligned with Research Question 4 as outlined in Section 3.1. The discussion focuses on five perspectives: multitasking, data, approach, transferability, and framework.

4.4.1. Multitask road traffic accident prediction

The literature shows that the existing RTA prediction models are

commonly designed to focus on performing one category of prediction at a time. To further advance the investigation in RTA prediction, a potential direction involves exploring the possibility of integrating multiple tasks into the RTA prediction models, such as the combination of the RTA risk level and occurrence. It is trusted to be a crucial direction because the integration helps increase the RTA prediction's effectiveness. A multitask prediction model provides more informative RTA notifications for road users or responsible authorities to take appropriate actions. Concurrently, it also reduces the efforts required to develop multiple different models. To support this claim, studies by Zhao et al. (2020) and Ashikuzzaman et al. (2021) adopted the RTA datasets from the UK open data portal to develop models for predicting RTA severity and risk. Both models are concluded with satisfactory performance where the PSO-ANN model has an accuracy of 0.853 to predict RTA severity, while the AUC of the CNN-RF model is 0.992 to predict RTA risk. The superior performance of these models provided valuable insights for future research to investigate the feasibility of developing a model to predict RTA risk and severity level concurrently.

4.4.2. Data quality

In terms of data quality, class imbalance remains a common issue among the RTA datasets. It requires careful consideration and proper data balancing techniques to prevent further deterioration of the model's performance. This is highlighted by Wang and Kim (2019) and J. Zhang et al. (2018), who utilized the imbalanced RTA datasets and developed RTA severity prediction models with an RF algorithm. Although the models outperform the other baseline models, their performance is inferior as compared to other studies that applied data balancing techniques such as SMOTE. However, the literature also points out that there is no one-size-fits-all solution to address the class imbalance issue and it highly depends on the application purpose and dataset characteristics. For example, Meng et al. (2018) discovered that the GBDT algorithm performs better by assigning a larger weight to the minority class, without applying the data balancing technique. This suggests the necessity to analyze the effects of different data balancing techniques on the model's performance in future research so that the best solution can be selected to suit the models.

Table 4

Comparison of strengths, limitations, and suggested use cases among the ML approaches.

Approaches	Strengths	Limitations	Suggested Use Cases
Statistical Regression	- The simplest structure and is easier to implement through mathematical modeling. - The coefficient values from the mathematical equation are useful for discovering the relationships and identifying key RTA factors.	The modeling equation captures only the linear relationships and might not be suitable for RTA datasets that often exhibit high-dimensional, non-linear, and complex relationships.	- Datasets with small sizes. - Handle simpler relationships among the RTA features. - For example, apply it to analyze RTA characteristics of datasets with fewer features.
Conventional ML	- Various algorithms exist to tackle a wide range of RTA prediction tasks in which the datasets exhibit non-linear relationships. - The ability to capture complex interactions between the RTA features is better than the statistical regression approach.	- Most of the algorithms have a set of hyperparameters and manual tuning can be time-consuming. - It has a limited ability to handle extremely complex and large RTA datasets as compared to the deep learning approach. - Offers interpretability for the prediction but the level of interpretability is lower than the statistical regression approach.	- Structured datasets with medium to large sizes. - Analyze complex and non-linear relationships among the RTA features. - Common approaches to handling large RTA datasets while offering prediction interpretability.
Deep Learning	- It has the best ability to extract complex and non-linear relationships as compared to statistical regression and conventional ML approaches. - There are various algorithms to better extract the specific features such as CNN for spatial analysis and LSTM for time series analysis of RTA patterns.	- It requires larger training datasets and extensive efforts for hyperparameter tuning to minimize overfitting. - Most of the algorithms are considered “black boxes” with limited interpretability and require additional techniques (such as SHAP values) to analyze the feature importance and identify crucial features.	- Datasets with exceptionally large sizes - Capture the complex patterns from the extremely large datasets for RTA predictions. - Applied for tasks with specific requirements, such as CNN for analyzing spatial correlations and LSTM for capturing temporal variations.
Hybrid	It is possible to achieve better performance as compared to other categories by combining the	The hybrid of multiple algorithms can increase the model's complexity, causing additional	- Ideal for predictions with multiple processing stages. - To address the need for

Table 4 (continued)

Approaches	Strengths	Limitations	Suggested Use Cases
	- strengths of different algorithms. - The flexibility of integrating different algorithms promotes a more efficient handling of complex RTA data.	- consumption of computation resources. - Requires careful consideration of the algorithms to combine because it might not always result in a better performance.	- additional algorithms to support the main algorithm. - For example, applying conventional ML for feature engineering, or ensemble the predictions of both conventional ML and deep learning models for boosted accuracy in road safety predictions.

4.4.3. Real-time application and ethical considerations

Another concern regarding RTA prediction models is their practical feasibility for real-time applications. While the predictive models assist in analyzing RTA characteristics and identifying key factors, it is also vital to consider their applicability for real-time implementation. This is important because a core value of developing RTA prediction models is to use them for instant prediction. An on-time prediction allows the responsible authorities to formulate effective response and recovery strategies during emergencies. Current literature has limited discussion on how the necessary RTA feature inputs can be collected for ad-hoc prediction. The non-visual solutions can be implemented with fewer resources compared to the visual solutions, but further research is recommended to develop a feasible framework for real-time implementation. This is vital to promote a wider adoption of ML-based solutions for improved road safety in the LMICs.

To promote the wider adoption of ML techniques to enhance road safety, a scalable and modular framework is expected to provide flexibility for all countries to implement based on their available resources. Specifically, high-income countries could leverage their existing traffic surveillance infrastructure to reduce expenses. Conversely, cost-effective solutions should be proposed to the LMICs as an alternative to the traffic surveillance system. In addition, cloud computing technologies are beneficial for creating a centralized platform for ML models to make RTA predictions. This reduces the need for high-performance hardware, which can be costly for extensive deployments along the roads.

With these considerations, a potential framework for implementing non-visual solutions in practice is presented in Fig. 5. For real-time data collection, countries with advanced traffic surveillance systems can adopt the available sensors to collect instantaneous traffic data. Well-calibrated sensors are expected to obtain accurate real-time traffic information such as traffic flow, volume, and occupancy. Nevertheless, this may not be feasible for worldwide implementation, particularly for some regions in the LMICs that cannot afford the installation and maintenance cost to implement a complete sensory system. As a replacement, insights can be gained from cases that used social media data for RTA prediction (Zhang et al., 2019b, 2019a; Z. Zhang et al., 2018). The idea of utilizing humans as “sensors” could be an alternative to sensors so that near real-time RTA data can be collected. This opens research opportunities to develop a smartphone-based mobile application as a standardized platform for the public to upload RTA data in crash scenarios. This is expected to be more efficient compared to traditional reporting through calls. In addition, multisource data fusion (Fig. 5) can be applied to collect RTA data in real time for RTA prediction. Specifically, information that is unlikely to change frequently, such

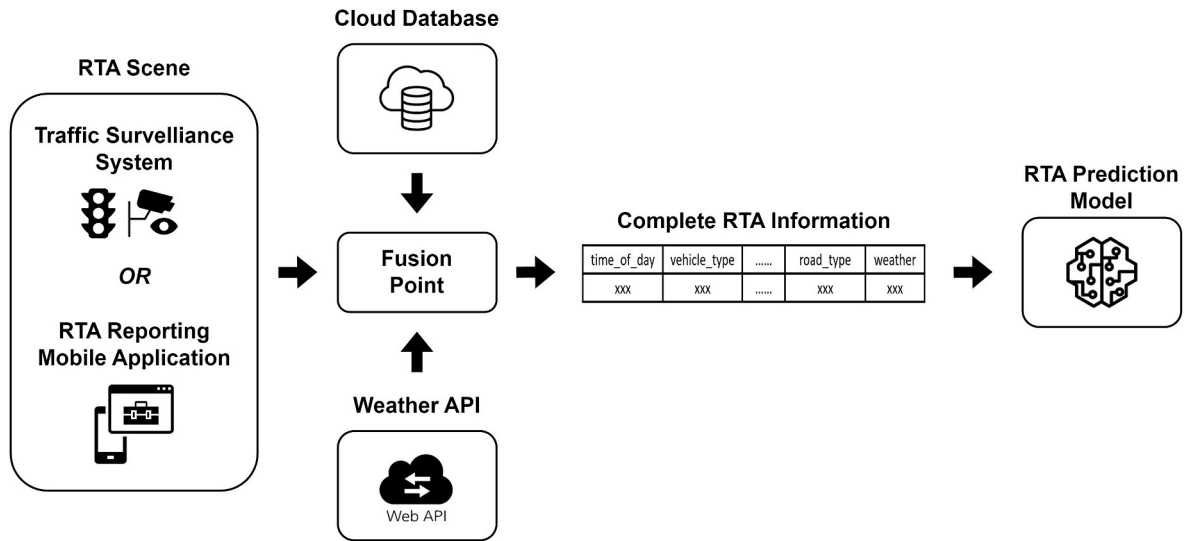


Fig. 5. Potential fusion-based framework for real-time implementation of non-visual RTA prediction solution.

as road geometry configurations and weather can be sourced through the cloud database and web application programming interface (API) based on the coordinates of the crash events. This is expected to speed up data collection processes by traffic surveillance systems or human reporters at the RTA scene, promoting a more effective implementation of the solutions.

There are also ethical concerns when advancing towards real-time ML-based RTA prediction with non-visual data. It can be analyzed from the perspectives of non-visual input collection and prediction outputs generated by ML models. Firstly, privacy concern emerges when developing solutions to collect non-visual inputs from various sources. Mobile applications can be useful for obtaining near real-time inputs, but there is a possibility that personal data (such as GPS location and personal details) could be collected and disclosed without consent from the road users (Boussada et al., 2023). The real implementation of non-visual RTA prediction solutions would require the collected data to be transferred to the cloud, introducing further concerns such as information leakage due to potential cybersecurity attacks. Thus, privacy protection is vital to ensure that no sensitive information is leaked when data is collected for intended operations (El Mestari et al., 2024). It is recommended to incorporate a proper mechanism so that the mobile application obtains consent before accessing and extracting sensitive data. Moreover, data collection should be aided with privacy-preserving ML schemes such as encryption, obfuscation, and aggregation (B. Liu et al., 2021). Eventually, all records need to be stored in secure storage with strict access authentication to mitigate data leakage.

Another issue is the bias in predicted outputs because the ML models are trained only on the characteristics of historical RTA cases. This is an example of representation bias (Mehrabi et al., 2021; van Giffen et al., 2022), where the training samples represent only a subset of all scenarios. For example, RTA occurrence and severity level often result from complex interactions between various factors that can vary over time due to changes in driver behaviors or environmental conditions. This introduces challenges for ML models to capture the relationships for perfectly accurate predictions, as witnessed in current literature, where none of the models achieved perfect performance for error-free predictions. To be ethically responsible, over-reliance on ML models for predictions should be avoided. Instead, monitoring plans and human supervision should be employed to mitigate the issue of representation bias when deploying non-visual solutions in real time (van Giffen et al., 2022). The predictions by ML models should serve as guidance during emergencies to analyze and understand scenarios for better formulation of action plans. Explainable AI techniques are also useful for the

emergency response team to understand and analyze the considerations made by the ML models. This allows a more balanced analysis by combining model interpretation with human oversight. Ultimately, ML models should be re-trained periodically to ensure they are updated with the latest RTA characteristics.

4.4.4. Hybrid approach for enhanced performance

Conventional ML is the predominant approach for RTA prediction with non-visual data. Specifically, the RF algorithm was investigated in 38 studies, with 16 of them concluding that it outperformed other baseline models. It is verified with the capability to handle high-dimensional RTA datasets through the ensemble of multiple DTs in parallel. Investigation with RF is considered to have reached its saturation point for RTA prediction, suggesting a shift in focus towards advanced techniques such as deep learning. Deep learning is reputed for its ability to handle extremely large datasets that exhibit complex correlations. However, most of the deep learning algorithms offer limited interpretability for their predictions. Considering this limitation, the hybrid approach emerges as a potential candidate to consider in future studies. It is expected to enhance the model's performance by combining the strengths of multiple algorithms while minimizing the weaknesses of the algorithms. Furthermore, it can also address the issue of limited interpretability faced by deep learning through the integration of interpretable algorithms such as statistical regression or conventional ML. To summarize, the superiority of the hybrid approach is a catalyst that motivates future research to advance the current knowledge in developing a more promising solution for RTA prediction with non-visual data.

Nonetheless, an observed limitation of the hybrid approach is that the model might have complex architecture and consume more computation resources. For a hybrid model, the computation resources are distributed among the algorithms that formed the model, resulting in a longer processing time. In particular, the LSTM-GBRT model proposed by Z. Zhang et al. (2020) has the best performance, but the slowest processing time among other baseline models. Moreover, similar findings are obtained by Yassin and Pooja (2020), where the proposed hybrid model consists of K-Means and RF takes more time to process the prediction of RTA severity level. Hence, a critical consideration in developing RTA prediction models with the hybrid approach is to obtain the best trade-off between performance and complexity. While hybrid models might offer better performance, they should have reasonable processing speed to make them feasible solutions for RTA prediction in real time.

4.4.5. Model transferability

Another important aspect of the RTA prediction models is the level of transferability. Several studies reviewed in this work demonstrate the applicability of developing RTA prediction models with various ML approaches to learn the RTA datasets from different regions. Although this is theoretically achievable, it could be challenging as multiple models are to be developed to handle the RTA characteristics that could be different across the regions. For example, [Rezapour et al. \(2020\)](#) and [Bedane et al. \(2021\)](#) developed prediction models to classify the severity level of RTAs in the US and Ethiopia, respectively. Both studies emphasize the significance of road characteristics and driver behaviors toward RTA severity levels. However, light condition is another key factor for Ethiopia which is not evident in the US case. These similarities and differences suggest the potential benefits of transfer learning for the rapid development of new models. The models can inherit the knowledge from pre-trained models and adapt to the specific RTA characteristics of the targeted region. This is proven by the DeepRisk framework from Zhang et al. (2019) to introduce a transferable solution for RTA risk estimation. Transfer learning technique is employed based on the social sensing data from four boroughs of New York City (Manhattan, Bronx, Queens, and Brooklyn) and achieved better RTA risk estimation with a reduction of MAE between 10.7% and 27.6%.

In summary, transfer learning helps to reduce the resources and research efforts needed as compared to developing a new model from scratch. Despite its potential, current literature shows limited discovery of transfer learning for RTA prediction with non-visual data. Therefore, a future research direction is to focus on improving the model development process by utilizing the transfer learning technique. While it is depicted to be an emerging opportunity, some critical issues are noteworthy. Firstly, the RTA datasets from different regions might not be recorded with the same columns or attributes. This poses a challenge in selecting the features, as the RTA features discarded by the pre-trained model might be significant for the target model. Additionally, it is vital to thoroughly consider the risk of negative transfer, where the target model is negatively affected by the knowledge learned from pre-trained models ([Zhuang et al., 2021](#)).

4.4.6. Standardized framework for model development

Analysis of the literature also revealed variations in the model development processes. The absence of a standardized framework causes the assessment of state-of-the-art progress to be a challenging task because different steps are involved in developing each model. To address this challenge, this study proposes a standardized framework as

shown in [Fig. 6](#) to serve as a reference for RTA prediction model development in future studies. The framework consists of 5 stages: (1) design ideation, (2) data collection, (3) data pre-processing, (4) prototyping, and (5) evaluation, which are summarized as follows:

- **Design Ideation:** At the design ideation stage, the primary concern is to define the characteristics of the RTA prediction task. Firstly, determining the scope is essential to set the overall direction of the investigation. This involves specifying the type of prediction task, which can be either CRM, COM, CFM, CSM or other RTA-related predictions. Another important aspect is to define the focus of the prediction task, whether it considers RTAs under a mixed traffic configuration, or targets specific road users such as pedestrians, two-wheelers, or heavy vehicles. Finalizing the model's input and output is also crucial because this influences the formulation of data collection strategy in the next stage.
- **Data Collection:** The model's input and output determined in the previous stage are essential to support the data collection stage. Findings from this study show that the common sources for RTA datasets include government departments, open-access databases, and sensory devices. Alternatively, simulation, social media, surveys, and web scraping can be considered when the common sources are unavailable. At the same time, each feature in the RTA datasets should be considered carefully on their availability in real-time to ensure the practicality of model deployment. For example, involving RTA features that cannot be collected instantaneously during the RTA might compromise the model's prediction performance.
- **Data Pre-processing:** After a suitable RTA dataset is sourced, pre-processing should be performed to assess data quality and prepare it into a form that can be fed to the RTA prediction model. This can be challenging due to the different issues and characteristics of individual RTA datasets. Therefore, this framework recommends three major tasks, namely noise (outliers) removal, missing values treatment, and class imbalance handling. The noise removal and missing value treatment are necessary to guarantee accurate correlations between the features by minimizing biases and maximizing information within the dataset. Moreover, effective class imbalance handling is also vital, especially for classification tasks such as classifying severity levels. This is because a predictive model trained with skewed data tends to overfit the majority class, resulting in inaccurate predictions of the minority class which often comprises fatal RTAs. For this case, data balancing techniques are essential so that an equal distribution of data is provided to the model to learn

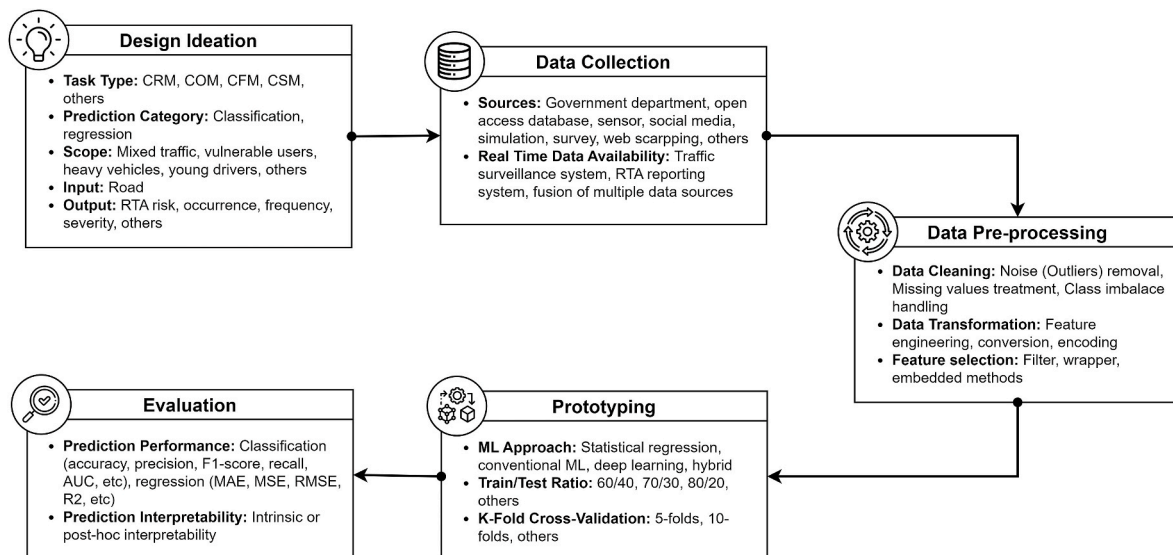


Fig. 6. The proposed framework to develop RTA prediction models.

and distinguish better between the severity levels. Another highlight of the pre-processing stage is the data transformation. Feature engineering is a frequent practice that involves the creation of new features based on existing features, such as deriving the time of day, day of week, and season information from crash occurrence time. Data transformation also includes formatting the RTA datasets into specific formats to fit the ML algorithms. This can involve the numerical-to-categorical conversion or encoding the data techniques such as label encoding and one-hot encoding.

- **Prototyping:** Following the pre-processing of collected RTA datasets, the development process continues with the prototyping to determine the main ML approach for building the model. A recommendation for ML approach selection is to align the dataset characteristics with the strengths of each ML approach. Statistical regression is suitable when the RTA features exhibit simplicity and mostly linear relationships. In contrast, conventional ML is the mainstream approach for RTA prediction with non-visual data especially for CSM, as observed in recent studies. Furthermore, deep learning and hybrid approaches are recognized for their flexibility to adapt to a variety of inputs and large-size datasets. In conclusion, it is advisable to thoroughly consider the expected characteristics of the models instead of making blindfold decisions to select the algorithms. In addition, decisions should also be made on the splitting of datasets into training and testing sets. Current literature shows that the split ratios typically ranged from 60% to 80% for training and 20%–40% for testing. To better estimate the performance, the k-fold cross-validation technique is recommended to validate the trained model with different data subsets.
- **Evaluation:** The final stage in the proposed framework involves verifying the applicability of the developed model to support RTA analysis and prediction. This is performed in two perspectives: prediction performance and interpretability. Performance evaluation depends on the nature of the prediction task, as identified during the design ideation stage. For example, a classification model is evaluated for its ability to accurately predict the class of targeted output, whereas the main concern for assessing the regression model is the errors between the estimation and actual values. As a result, different sets of metrics are used to evaluate the classification and regression models. For prediction interpretability, it includes the analysis to assess the level of transparency for the model's predictions. Statistical regression and conventional ML algorithms usually offer interpretable predictions where the feature coefficients or feature attribution outputs can be used for RTA factor analysis. Conversely, most deep learning algorithms offer limited interpretability and are frequently regarded as “black boxes.” Thus, incorporating external techniques is necessary to explain the predictions and analyze significant factors. To perform this, some explainable AI techniques applied in recent literature include the Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME).

5. Conclusion

The application of predictive models is useful to support various tasks in the transportation sector. From the perspective of RTA, it is especially vital for the LMICs which have higher fatality rates to assist in the prevention, response, and recovery of RTA. While technology constantly advances and changes, this study comprehensively reviews the state-of-the-art solutions to investigate the effects of advanced technology such as ML, on its effectiveness for RTA prediction. This is essential to identify the current trends and suggest future directions to improve the modeling and analysis of RTA data. Moreover, this study emphasizes the RTA prediction with non-visual data rather than traffic images or video data because it is deemed more feasible for implementation in all regions worldwide including the LMICs.

Firstly, the non-visual RTA datasets can be obtained from various sources, where existing studies showed that government departments

and open-access data portals are the commonly used sources. Furthermore, recent studies also demonstrated the maturity and prominence of conventional ML as compared to statistical regression, deep learning, and hybrid approaches. Although statistical regression is claimed to have the simplest architecture and offer high interpretability, it is not suitable for RTA datasets that contain complex patterns and relationships. In contrast, although deep learning and hybrid approaches are not widely investigated, they are still worth more discovery in future research due to their scalability and flexibility. Remarkably, the hybrid approach offers the potential to further advance research in this field by investigating the integration of different algorithms to maximize their respective strengths. This can be applied to support the selection of optimal RTA features or tune the main algorithm with the most optimum structure.

The discussion and critical analysis in this study also sparked some potential directions to consider in future research. Firstly, future research is recommended to develop multitask predictive models to perform several RTA prediction tasks with only one model. There are solutions from recent studies with satisfactory performance in performing one RTA prediction task at a time. Therefore, investigating the possibility of integrating multiple tasks into a model can help to reduce the consumption of computation resources and minimize processing time. As for the RTA data sources, a fusion-based data collection concept is proposed for enhanced feasibility of the future solution. With this concept, RTA data is collected from multiple sources and combined into an informative input for RTA predictions, allowing a faster collection of RTA data at the crash scene. Besides, this study has witnessed the potential of hybrid approach to develop more flexible models. Its investigation has not reached the maturity point as compared to conventional ML approaches and thus, a suggested direction to focus on in future research. Moreover, transfer learning is another technique that is worth investigating. There are differences in RTA characteristics around the globe. Therefore, transfer learning can be used for a model to adapt to the changes in RTA characteristics, instead of developing multiple models from scratch. Although transfer learning is challenging to implement, it is believed to be a resource-efficient solution that reduces the efforts in developing RTA prediction models for different regions. Lastly, this study also conceptualized a standard framework that serves as guidelines for RTA prediction model development. With this framework, it is possible to conduct a more systematic evaluation of the performance and effectiveness of each solution in the future.

CRedit authorship contribution statement

Albe Bing Zhe Chai: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Bee Theng Lau:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing. **Mark Kit Tsun Tee:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Chris McCarthy:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This research is fully funded by the Swinburne HDR Scholarship.

References

- Abdulrashid, I., Zanjirani Farahani, R., Mammadov, S., Khalafalla, M., Chiang, W.-C., 2024. Explainable artificial intelligence in transport Logistics: risk analysis for road accidents. *Transp Res E Logist Transp Rev* 186, 103563. <https://doi.org/10.1016/j.tre.2024.103563>.
- Abouloula, O.I., 2024. Improving traffic accident severity prediction using MobileNet transfer learning model and SHAP XAI technique. *PLoS One* 19, e0300640. <https://doi.org/10.1371/JOURNAL.PONE.0300640>.
- Almamlook, R.E., Kwayu, K.M., Alkasisbeh, M.R., Frefer, A.A., 2019. Comparison of machine learning algorithms for predicting traffic accident severity. In: 2019 IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology, JEEIT 2019 - Proceedings. Institute of Electrical and Electronics Engineers Inc., pp. 272–276. <https://doi.org/10.1109/JEEIT.2019.8717393>.
- Alnami, H.M., Mahgoub, I., Al-Najada, H., 2021. Highway accident severity prediction for optimal resource allocation of emergency vehicles and personnel. In: 2021 IEEE 11th Annual Computing and Communication Workshop and Conference, CCWC 2021. Institute of Electrical and Electronics Engineers Inc., pp. 1231–1238. <https://doi.org/10.1109/CCWC51732.2021.9376155>.
- Alqatawna, A., Rivas Alvarez, A.M., Garcia-Moreno, S.S.C., 2021. Comparison of multivariate regression models and artificial neural networks for prediction highway traffic accidents in Spain: a case study. *Transport. Res. Procedia* 58, 277–284. <https://doi.org/10.1016/j.trpro.2021.11.038>.
- Al-Radaideh, Q.A., Daoud, E.J., 2018. Data mining methods for traffic accident severity prediction. *International Journal of Neural Networks and Advanced Applications* 5, 1–12.
- Amiri, A.M., Sadri, A., Nadimi, N., Shams, M., 2020. A comparison between Artificial Neural Network and Hybrid Intelligent Genetic Algorithm in predicting the severity of fixed object crashes among elderly drivers. *Accid. Anal. Prev.* 138, 105468. <https://doi.org/10.1016/j.aap.2020.105468>.
- Ashikuzzaman, M., Akram, W., Mydul Islam Anik, M., Hasan, M., Sawkat Ali, M., Jabid, T., 2021. PSO-ANN in preventing traffic collisions: a comparative study. *Indonesian Journal of Electrical Engineering and Computer Science* 24, 1796–1803. <https://doi.org/10.11591/IJEECS.V24.I3.PP1796-1803>.
- Assi, K., 2020. Traffic crash severity prediction—a synergy by hybrid principal component analysis and machine learning models. *Int. J. Environ. Res. Publ. Health* 17, 7598. <https://doi.org/10.3390/IJERPH1707598>.
- Assi, K., Rahman, S.M., Mansoor, U., Ratrou, N., 2020. Predicting crash injury severity with machine learning algorithm synergized with clustering technique: a promising protocol. *Int. J. Environ. Res. Publ. Health* 17, 5497. <https://doi.org/10.3390/IJERPH17155497>.
- Bahiru, T.K., Kumar Singh, D., Tessfaw, E.A., 2018. Comparative study on data mining classification algorithms for predicting road traffic accident severity. In: Proceedings of the International Conference on Inventive Communication and Computational Technologies, ICICCT 2018. Institute of Electrical and Electronics Engineers Inc., pp. 1655–1660. <https://doi.org/10.1109/ICICCT.2018.8473265>.
- Balawi, M., Tenekci, G., 2024. Time series traffic collision analysis of London hotspots: patterns, predictions and prevention strategies. *Heliyon* 10, e25710. <https://doi.org/10.1016/j.heliyon.2024.e25710>.
- Basso, F., Basso, L.J., Bravo, F., Pezoa, R., 2018. Real-time crash prediction in an urban expressway using disaggregated data. *Transport. Res. C Emerg. Technol.* 86, 202–219. <https://doi.org/10.1016/j.trc.2017.11.014>.
- Bedane, T.T., Assefa, B.G., Mohapatra, S.K., 2021. Preventing traffic accidents through machine learning predictive models. In: 2021 International Conference on Information and Communication Technology for Development for Africa, ICT4DA 2021. Institute of Electrical and Electronics Engineers Inc., pp. 36–41. <https://doi.org/10.1109/ICT4DA53266.2021.9672249>.
- Beryl Princess, P.J., Silas, S., Rajasingh, E.B., 2021. Performance comparison of machine learning models for classification of traffic injury severity from imbalanced accident dataset. In: Advances in Intelligent Systems and Computing. Springer, pp. 361–369. https://doi.org/10.1007/978-981-15-5285-4_36.
- Bhardwaj, B.B., Goswami, S., 2019. Road asset management system in developing countries-implementation and challenges. *Int. J. Civ. Eng. Technol.* 10, 1227–1232.
- Boussada, R., Bouchaala, M., Saidane, L.A., 2023. Privacy and tracking in the emerging mobile applications: a survey. 2023 International Wireless Communications and Mobile Computing, IWCMC 2023, 1491–1496. <https://doi.org/10.1109/IWCMC58020.2023.10182497>.
- Cai, Q., Abdel-Aty, M., Yuan, J., Lee, J., Wu, Y., 2020. Real-time crash prediction on expressways using deep generative models. *Transport. Res. C Emerg. Technol.* 117, 102697. <https://doi.org/10.1016/j.trc.2020.102697>.
- Chakraborty, A., Mukherjee, D., Mitra, S., 2019. Development of pedestrian crash prediction model for a developing country using artificial neural network. *Int. J. Inj. Control Saf. Promot.* 26, 283–293. <https://doi.org/10.1080/17457300.2019.1627463>.
- Chen, X., 2021. Traffic crash severity prediction with deep learning. *J Phys Conf Ser* 1883, 012141. <https://doi.org/10.1088/1742-6596/1883/1/012141>.
- Danesh, A., Ehsani, M., Moghadas Nejad, F., Zakeri, H., 2022. Prediction model of crash severity in imbalanced dataset using data leveling methods and metaheuristic optimization algorithms. *Int. J. Crashworthiness* 27, 1869–1882. <https://doi.org/10.1080/13588265.2022.2028471>.
- Das, S., Le, M., Dai, B., 2020. Application of machine learning tools in classifying pedestrian crash types: a case study. *Transportation Safety and Environment* 2, 106–119. <https://doi.org/10.1093/TSE/TDA010>.
- Dia, Y., Faty, L., Sarr, M.D., Sall, O., Bouso, M., Landu, T.T., 2022. Study of supervised learning algorithms for the prediction of road accident severity in Senegal. In: 2022 7th International Conference on Computational Intelligence and Applications, ICCIA 2022. Institute of Electrical and Electronics Engineers Inc., pp. 123–127. <https://doi.org/10.1109/ICCIA55271.2022.9828434>.
- Dogru, N., Subasi, A., 2018. Traffic accident detection using random forest classifier. In: 2018 15th Learning and Technology Conference, L and T 2018. Institute of Electrical and Electronics Engineers Inc., pp. 40–45. <https://doi.org/10.1109/LT.2018.8368509>.
- Donchenko, D., Sadovnikova, N., Parygin, D., 2020. Prediction of road accidents' severity on Russian roads using machine learning techniques. In: Proceedings of the 5th International Conference on Industrial Engineering (ICIE 2019). Springer, pp. 1493–1501. https://doi.org/10.1007/978-3-030-22063-1_157.
- Dong, C., Shao, C., Li, J., Xiong, Z., 2018. An improved deep learning model for traffic crash prediction. *J. Adv. Transport.* <https://doi.org/10.1155/2018/3869106>.
- El Mestari, S.Z., Lenzini, G., Demirci, H., 2024. Preserving data privacy in machine learning systems. *Comput. Secur.* 137, 103605. <https://doi.org/10.1016/J.COSE.2023.103605>.
- Elamrani Abou Elassad, Z., Mousannif, H., Al Moatassime, H., 2020. Class-imbalanced crash prediction based on real-time traffic and weather data: a driving simulator study. *Traffic Inj. Prev.* 21, 201–208. <https://doi.org/10.1080/15389588.2020.1723794>.
- Emu, M., Kamal, F.B., Choudhury, S., Rahman, Q.A., 2022. Fatality prediction for motor vehicle collisions: mining big data using deep learning and ensemble methods. *IEEE Open Journal of Intelligent Transportation Systems* 3, 199–209. <https://doi.org/10.1109/OJITS.2022.3160404>.
- Fiorentini, N., Losa, M., 2020. Handling imbalanced data in road crash severity prediction by machine learning algorithms. *Infrastructure* 5, 61. <https://doi.org/10.3390/INFRASTRUCTURES5070061>.
- Ghasedi, M., Sarfjo, M., Bargogol, I., 2021. Prediction and analysis of the severity and number of suburban accidents using Logit model, factor analysis and machine learning: a case study in a developing country. *SN Appl. Sci.* 3, 13. <https://doi.org/10.1007/s42452-020-04081-3>.
- Gu, X., Li, T., Wang, Yonghui, Zhang, L., Wang, Yitian, Yao, J., 2018. Traffic fatalities prediction using support vector machine with hybrid particle swarm optimization. *J. Algorithm Comput. Technol.* 12, 20–29. <https://doi.org/10.1177/1748301817729953>.
- Guo, M., Janson, B., Peng, Y., 2024. A spatiotemporal deep learning approach for pedestrian crash risk prediction based on POI trip characteristics and pedestrian exposure intensity. *Accid. Anal. Prev.* 198, 107493. <https://doi.org/10.1016/J.AAP.2024.107493>.
- Gutierrez-Osorio, C., Pedraza, C., 2020. Modern data sources and techniques for analysis and forecast of road accidents: a review. *J. Traffic Transport. Eng.* 7, 432–446. <https://doi.org/10.1016/J.JTTE.2020.05.002>.
- Hadjimitsi, N.S., Lippi, M., Dell'Amico, M., Skiera, A., 2020. Machine learning for severity classification of accidents involving powered two wheelers. *IEEE Trans. Intell. Transport. Syst.* 21, 4308–4317. <https://doi.org/10.1109/TITS.2019.2939624>.
- Haruna, L., Sallehuddin, R., Radzi, H.M., 2020. Discrete particle swarm optimization based filter feature selection technique for the severity of road traffic accident prediction. In: Advances in Intelligent Systems and Computing. Springer, pp. 298–310. https://doi.org/10.1007/978-3-030-33582-3_29.
- Hasan, A.S., Jalayer, M., Das, S., Asif Bin Kabir, M., 2023. Application of machine learning models and SHAP to examine crashes involving young drivers in New Jersey. *International Journal of Transportation Science and Technology.* <https://doi.org/10.1016/J.IJTST.2023.04.005>.
- Hasheminejad, S.H.A., Zahedi, M., Hasheminejad, S.M.H., 2018. A hybrid clustering and classification approach for predicting crash injury severity on rural roads. *Int. J. Inj. Control Saf. Promot.* 25, 85–101. <https://doi.org/10.1080/17457300.2017.1341933>.
- Huang, T., Wang, S., Sharma, A., 2020. Highway crash detection and risk estimation using deep learning. *Accid. Anal. Prev.* 135, 105392. <https://doi.org/10.1016/J.AAP.2019.105392>.
- Hyder, A.A., Hoe, C., Hajar, M., Peden, M., 2022. The political and social contexts of global road safety: challenges for the next decade. *Lancet* 400, 127–136. [https://doi.org/10.1016/S0140-6736\(22\)00917-5](https://doi.org/10.1016/S0140-6736(22)00917-5).
- Ijaz, M., Ian, L., Zahid, M., Jamal, A., 2021. A comparative study of machine learning classifiers for injury severity prediction of crashes involving three-wheeled motorized rickshaw. *Accid. Anal. Prev.* 154, 106094. <https://doi.org/10.1016/J.AAP.2021.106094>.
- Infante, P., Jacinto, G., Afonso, A., Rego, L., Nogueira, V., Quaresma, P., Saias, J., Santos, D., Nogueira, P., Silva, M., Costa, R.P., Gois, P., Manuel, P.R., 2022. Comparison of statistical and machine-learning models on road traffic accident severity classification. *Computers* 11, 80. <https://doi.org/10.3390/COMPUTERS11050080>.
- Jeong, H., Jang, Y., Bowman, P.J., Masoud, N., 2018. Classification of motor vehicle crash injury severity: a hybrid approach for imbalanced data. *Accid. Anal. Prev.* 120, 250–261. <https://doi.org/10.1016/J.AAP.2018.08.025>.
- Jianfeng, X., Hongyu, G., Jian, T., Liu, L., Haizhu, L., 2019. A classification and recognition model for the severity of road traffic accident. *Adv. Mech. Eng.* 11, 168781401985189. <https://doi.org/10.1177/1687814019851893>.
- Jiang, F., Yuen, K.K.R., Lee, E.W.M., 2020. A long short-term memory-based framework for crash detection on freeways with traffic data of different temporal resolutions. *Accid. Anal. Prev.* 141, 105520. <https://doi.org/10.1016/J.AAP.2020.105520>.
- Kashifi, M.T., Al-Turki, M., Sharify, A.W., 2023. Deep hybrid learning framework for spatiotemporal crash prediction using big traffic data. *International Journal of Transportation Science and Technology* 12, 793–808. <https://doi.org/10.1016/J.IJTST.2022.07.003>.

- Katanalp, B.Y., Eren, E., 2020. The novel approaches to classify cyclist accident injury-severity: hybrid fuzzy decision mechanisms. *Accid. Anal. Prev.* 144, 105590 <https://doi.org/10.1016/j.aap.2020.105590>.
- Khan, M.N., Das, A., Ahmed, M.M., 2024. Prediction of truck-involved crash severity on a rural mountainous freeway using transfer learning with ResNet-50 deep neural network. *J. Transp. Eng. A Syst.* 150, 04023131. <https://doi.org/10.1061/JTEPBS.TEENG-7304>.
- Kitchenham, B.A., 2004. *Procedures for Performing Systematic Reviews*. Keele.
- Kumeda, B., Zhang, F., Zhou, F., Hussain, S., Almasri, A., Assefa, M., 2019. Classification of road traffic accident data using machine learning Algorithms. In: 2019 IEEE 11th International Conference on Communication Software and Networks, ICCSN 2019. Institute of Electrical and Electronics Engineers Inc., pp. 682–687. <https://doi.org/10.1109/ICCSN.2019.8905362>.
- Labib, M.F., Rifat, A.S., Hossain, M.M., Das, A.K., Nawrine, F., 2019. Road accident analysis and prediction of accident severity by using machine learning in Bangladesh. In: 2019 7th International Conference on Smart Computing and Communications, ICSCC 2019. Institute of Electrical and Electronics Engineers Inc. <https://doi.org/10.1109/ICSCC.2019.8843640>.
- Lee, Jonghak, Yoon, T., Kwon, S., Lee, Jongtae, 2020. Model evaluation for forecasting traffic accident severity in rainy seasons using machine learning algorithms: seoul city study. *Appl. Sci.* 10, 129. <https://doi.org/10.3390/AP10010129>.
- Li, P., Abdel-Aty, M., Yuan, J., 2020. Real-time crash risk prediction on arterials based on LSTM-CNN. *Accid. Anal. Prev.* 135, 105371 <https://doi.org/10.1016/j.aap.2019.105371>.
- Lin, C., Wu, D., Liu, H., Xia, X., Bhattarai, N., 2020. Factor identification and prediction for teen driver crash severity using machine learning: a case study. *Appl. Sci.* 10, 1675. <https://doi.org/10.3390/AP10051675>.
- Liu, B., Ding, M., Shaham, S., Rahayu, W., Farokhi, F., Lin, Z., 2021. When machine learning meets privacy. *ACM Comput. Surv.* 54, 31. <https://doi.org/10.1145/3436755>.
- Liu, J., Boyle, L.N., Banerjee, A.G., 2018. Predicting interstate motor carrier crash rate level using classification models. *Accid. Anal. Prev.* 120, 211–218. <https://doi.org/10.1016/j.aap.2018.06.005>.
- Liu, R., Zhang, L., Wang, J., Yang, H., Liu, Y., 2021. PETRI: reducing bandwidth requirement in smart surveillance by edge-cloud collaborative adaptive frame clustering and pipelined bidirectional tracking. In: *Proceedings - Design Automation Conference*. Institute of Electrical and Electronics Engineers Inc., pp. 421–426. <https://doi.org/10.1109/DAC18074.2021.9586088>.
- Mamo, D.E., Abebe, A., Beyene, T., Alemu, F., Bereka, B., 2023. Road traffic accident clinical pattern and management outcomes at JUMC Emergency Department; Ethiopia. *African Journal of Emergency Medicine* 13, 1–5. <https://doi.org/10.1016/J.AFJEM.2022.11.002>.
- Marcello, P., Valdivieso Caraguay, Á.L., Hernández-álvarez, M., 2022. A systematic literature review of learning-based traffic accident prediction models based on heterogeneous sources. *Appl. Sci.* 12, 4529. <https://doi.org/10.3390/AP12094529>.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., Galstyan, A., 2021. A Survey on Bias and Fairness in Machine Learning, 54. *ACM Computing Surveys (CSUR)*. <https://doi.org/10.1145/3457607>.
- Meng, H., Wang, Xinhong, Wang, Xuesong, 2018. Expressway crash prediction based on traffic big data. In: *ACM International Conference Proceeding Series*. Association for Computing Machinery, pp. 11–16. <https://doi.org/10.1145/3297067.3297093>.
- Mo, W., Lee, J., Abdel-Aty, M., Mao, S., Jiang, Q., 2024. Dynamic short-term crash analysis and prediction at toll plazas for proactive safety management. *Accid. Anal. Prev.* 197, 107456 <https://doi.org/10.1016/j.aap.2024.107456>.
- Mohanty, M., Panda, R., Gandupalli, S.R., Sonowal, D., Muskan, M., Chakraborty, R., Dangeti, M.R., 2022. Development of crash prediction models by assessing the role of perpetrators and victims: a comparison of ANN & logistic model using historical crash data. *Int. J. Inj. Control Saf. Promot.* <https://doi.org/10.1080/17457300.2022.2089899>.
- Mokhtarimousavi, S., Anderson, J.C., Aziznamini, A., Hadi, M., 2019. Improved support vector machine models for work zone crash injury severity prediction and analysis. *Transport. Res. Rec.: J. Transport. Res. Board* 2673, 680–692. <https://doi.org/10.1177/0361198119845899>.
- Mokoatle, M., Marivate, V., Esiefariene, M., 2019. Predicting road traffic accident severity using accident report data in South Africa. In: *ACM International Conference Proceeding Series*, pp. 11–17. <https://doi.org/10.1145/3325112.3325211>.
- Mrema, I.J., Dida, M.A., 2020. A survey of road accident reporting and driver's behavior awareness systems: the case of Tanzania. *Eng. Technol. Appl. Sci. Res.* 10, 6009–6015.
- Nassiri, H., Mohammadpour, S.I., Dahaghin, M., 2022. Forecasting time trends of fatal motor vehicle crashes in Iran using an ensemble learning algorithm. *Traffic Inj. Prev.* 24, 44–49. <https://doi.org/10.1080/15389588.2022.2130279>.
- Nguyen, H., Kieu, L.M., Wen, T., Cai, C., 2018. Deep learning methods in transportation domain: a review. *IET Intell. Transp. Syst.* 12, 998–1004. <https://doi.org/10.1049/IET-ITS.2018.0064>.
- Panda, C., Mishra, A.K., Dash, A.K., Nawab, H., 2022. Predicting and explaining severity of road accident using artificial intelligence techniques, SHAP and feature analysis. *Int. J. Crashworthiness* 28, 186–201. <https://doi.org/10.1080/13588265.2022.2074643>.
- Panicker, A.K., Ramadurai, G., 2020. Injury severity prediction model for two-wheeler crashes at mid-block road sections. *Int. J. Crashworthiness* 27, 328–336. <https://doi.org/10.1080/13588265.2020.1806644>.
- Parsa, A.B., Taghipour, H., Derrible, S., Mohammadian, A.Kouros, 2019. Real-time accident detection: coping with imbalanced data. *Accid. Anal. Prev.* 129, 202–210. <https://doi.org/10.1016/j.aap.2019.05.014>.
- Rahim, M.A., Hassan, H.M., 2021. A deep learning based traffic crash severity prediction framework. *Accid. Anal. Prev.* 154, 106090 <https://doi.org/10.1016/j.aap.2021.106090>.
- Raul, S.K., Rout, R.R., Somayajulu, D.V.L.N., 2024. Weighted ensemble learning for accident severity classification using social media data. *SN Comput. Sci.* 5, 1–20. <https://doi.org/10.1007/S42979-024-02870-W>.
- Reichenbach, A., Navarro, B., E. J., 2021. A model for traffic incident prediction using emergency braking data. In: *IEEE Intelligent Vehicles Symposium, Proceedings*. Institute of Electrical and Electronics Engineers Inc., pp. 22–27. <https://doi.org/10.1109/IV48863.2021.9575200>.
- Ren, H., Song, Y., Wang, J., Hu, Y., Lei, J., 2018. A deep learning approach to the citywide traffic accident risk prediction. In: *IEEE Conference on Intelligent Transportation Systems, Proceedings, ITSC*. Institute of Electrical and Electronics Engineers Inc., pp. 3346–3351. <https://doi.org/10.1109/ITSC.2018.8569437>.
- Rezapour, M., Mehrara Molan, A., Ksaibati, K., 2020. Analyzing injury severity of motorcycle at-fault crashes using machine learning techniques, decision tree and logistic regression models. *International Journal of Transportation Science and Technology* 9, 89–99. <https://doi.org/10.1016/J.IJTST.2019.10.002>.
- Sarkar, A., Sarkar, S., 2020. Comparative assessment between statistical and soft computing methods for accident severity classification. *J. Inst. Eng.: Series A* 101, 27–40. <https://doi.org/10.1007/s40030-019-00422-7>.
- Sattar, K., Chikh Oughali, F., Assi, K., Ratroun, N., Jamal, A., Masiur Rahman, S., 2023. Transparent deep machine learning framework for predicting traffic crash severity. *Neural Comput. Appl.* 35, 1535–1547. <https://doi.org/10.1007/s00521-022-07769-2>.
- Shao, Y., Shi, X., Zhang, Y., Shiwakoti, N., Xu, Y., Ye, Z., 2024. Injury severity prediction and exploration of behavior-cause relationships in automotive crashes using natural language processing and extreme gradient boosting. *Eng. Appl. Artif. Intell.* 133, 108542 <https://doi.org/10.1016/J.ENGAPAI.2024.108542>.
- Shetty, R.R., Liu, H., 2021. Accidents analysis and severity prediction using machine learning algorithms. In: *AI and Analytics for Smart Cities and Service Systems, INFORMS International Conference on Service Science (ICSS 2021)*. Springer, Cham, pp. 173–183. https://doi.org/10.1007/978-3-030-90275-9_15.
- Shi, L., Qian, C., Guo, F., 2022. Real-time driving risk assessment using deep learning with XGBoost. *Accid. Anal. Prev.* 178, 106836 <https://doi.org/10.1016/J.AAP.2022.106836>.
- Siddik, MdA.B., Arman, MdS., Hasan, A., Jahan, M.R., Islam, M., Biplob, K.B.B., 2021. Predicting the death of road accidents in Bangladesh using machine learning algorithms. In: *Communications in Computer and Information Science*. Springer Science and Business Media Deutschland GmbH, pp. 160–171. https://doi.org/10.1007/978-3-030-88244-0_16.
- Silva, P.B., Andrade, M., Ferreira, S., 2020. Machine learning applied to road safety modeling: a systematic literature review. *J. Traffic Transport. Eng.* <https://doi.org/10.1016/j.jtte.2020.07.004>.
- Singh, G., Pal, M., Yadav, Y., Singla, T., 2020. Deep neural network-based predictive modeling of road accidents. *Neural Comput. Appl.* 32, 12417–12426. <https://doi.org/10.1007/S00521-019-04695-8>.
- Singh, G., Sachdeva, S.N., Pal, M., 2018. Support vector machine model for prediction of accidents on non-urban sections of highways. *Transport* 171, 253–263. <https://doi.org/10.1680/JTRAN.17.00019>.
- Tao, W., Aghaabbasi, M., Ali, M., Almaliki, A.H., Zainol, R., Almaliki, A.A., Hussein, E.E., 2022. An advanced machine learning approach to predicting pedestrian fatality caused by road crashes: a step toward sustainable pedestrian safety. *Sustainability* 14, 2436. <https://doi.org/10.3390/SU14042436>.
- Tariq, A.-M., Xiao, H., Jean, P.N., Eshrak, N.A., Ibrahim, A., 2020. Exploiting machine learning algorithms for predicting crash injury severity in Yemen: hospital case study. *Appl. Comput. Math.* 9, 164. <https://doi.org/10.11648/J.ACM.20200905.12>.
- Tay, H.L., Banomyong, R., Varadejsatitwong, P., Julagasigorn, P., 2022. Mitigating risks in the disaster management cycle. *Adv. Civ. Eng.* 1–14. <https://doi.org/10.1155/2022/7454760>.
- Theofilatos, A., Chen, C., Antoniou, C., 2019. Comparing machine learning and deep learning methods for real-time crash prediction. *Transport. Res. Rec.: J. Transport. Res. Board* 2673, 169–178. <https://doi.org/10.1177/0361198119841571>.
- Vadhwani, D., Thakor, D., 2022. Prediction of extent of damage in vehicle during crash using improved XGBoost model. *Int. J. Crashworthiness*. <https://doi.org/10.1080/13588265.2022.2075101>.
- van Giffen, B., Herhausen, D., Fahse, T., 2022. Overcoming the pitfalls and perils of algorithms: a classification of machine learning biases and mitigation methods. *J. Bus. Res.* 144, 93–106. <https://doi.org/10.1016/J.JBUSRES.2022.01.076>.
- Wahab, L., Jiang, H., 2020. Severity prediction of motorcycle crashes with machine learning methods. *Int. J. Crashworthiness* 25, 485–492. <https://doi.org/10.1080/13588265.2019.1616885>.
- Wahab, L., Jiang, H., 2019. A comparative study on machine learning based algorithms for prediction of motorcycle crash severity. *PLoS One* 14, e0214966. <https://doi.org/10.1371/JOURNAL.PONE.0214966>.
- Wang, J., Song, H., Fu, T., Behan, M., Jie, L., He, Y., Shangquan, Q., 2022. Crash prediction for freeway work zones in real time: a comparison between Convolutional Neural Network and Binary Logistic Regression model. *International Journal of Transportation Science and Technology* 11, 484–495. <https://doi.org/10.1016/J.IJTST.2021.06.002>.
- Wang, X., Kim, S.H., 2019. Prediction and factor identification for crash severity: comparison of discrete choice and tree-based models. *Transport. Res. Rec.: J. Transport. Res. Board* 2673, 640–653. <https://doi.org/10.1177/0361198119844456>.
- Wang, Z., Jiao, P., Wang, J., Huang, Q., Li, R., Lu, H., 2022. The level of delay caused by crashes (LDC) in metropolitan and non-metropolitan areas: a comparative analysis of

- improved Random Forests and LightGBM. *Int. J. Crashworthiness*. <https://doi.org/10.1080/13588265.2022.2130624>.
- WHO, 2022. Road Traffic Injuries [WWW Document]. URL. <https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>, 11.22.22.
- WHO, 2018. GLOBAL STATUS REPORT ON ROAD SAFETY 2018 SUMMARY. World Health Organization.
- Wu, D., Wang, S., 2020. Comparison of road traffic accident prediction effects based on SVR and BP neural network. In: *Proceedings of 2020 IEEE International Conference on Information Technology, Big Data and Artificial Intelligence, ICIBA 2020*. Institute of Electrical and Electronics Engineers Inc., pp. 1150–1154. <https://doi.org/10.1109/ICIBA50161.2020.9277150>
- Xi, J., Guo, H., Tian, J., Liu, L., Liu, H., 2019. A classification and recognition model for the severity of road traffic accident. *Adv. Mech. Eng.* 11 <https://doi.org/10.1177/1687814019851893>, 168781401985189.
- Yahaya, M., Jiang, X., Fu, C., Bashir, K., Fan, W., 2019. Enhancing crash injury severity prediction on imbalanced crash data by sampling technique with variable selection. In: *2019 IEEE Intelligent Transportation Systems Conference, ITSC 2019*. Institute of Electrical and Electronics Engineers Inc., pp. 363–368. <https://doi.org/10.1109/ITSC.2019.8917223>
- Yang, K., Quddus, M., Antoniou, C., 2022. Developing a new real-time traffic safety management framework for urban expressways utilizing reinforcement learning tree. *Accid. Anal. Prev.* 178, 106848 <https://doi.org/10.1016/J.AAP.2022.106848>.
- Yassin, S.S., Pooja, 2020. Road accident prediction and model interpretation using a hybrid K-means and random forest algorithm approach. *SN Appl. Sci.* 2, 1–13. <https://doi.org/10.1007/S42452-020-3125-1/TABLES/5>.
- Zhang, C., He, J., Wang, Y., Yan, X., Zhang, Changjian, Chen, Y., Liu, Z., Zhou, B., 2020. A crash severity prediction method based on improved neural network and factor analysis. *Discrete Dynam Nat. Soc.* 1–13. <https://doi.org/10.1155/2020/4013185>.
- Zhang, J., Li, Z., Pu, Z., Xu, C., 2018. Comparing prediction performance for crash injury severity among various machine learning and statistical methods. *IEEE Access* 6, 60079–60087. <https://doi.org/10.1109/ACCESS.2018.2874979>.
- Zhang, Y., Li, H., Ren, G., 2023. Analyzing the injury severity in single-bicycle crashes: an application of the ordered forest with some practical guidance. *Accid. Anal. Prev.* 189, 107126 <https://doi.org/10.1016/J.AAP.2023.107126>.
- Zhang, Y., Wang, H., Zhang, D., Lu, Y., Wang, D., 2019a. RiskCast: social sensing based traffic risk forecasting via inductive multi-view learning. In: *2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. Association for Computing Machinery, Inc, pp. 154–157. <https://doi.org/10.1145/3341161.3342912>.
- Zhang, Y., Wang, H., Zhang, D., Wang, D., 2019b. DeepRisk: a deep transfer learning approach to migratable traffic risk estimation in intelligent transportation using social sensing. In: *Proceedings - 15th Annual International Conference on Distributed Computing in Sensor Systems, DCOSS 2019*. Institute of Electrical and Electronics Engineers Inc., pp. 123–130. <https://doi.org/10.1109/DCOSS.2019.00039>
- Zhang, Z., He, Q., Gao, J., Ni, M., 2018. A deep learning approach for detecting traffic accidents from social media data. *Transport. Res. C Emerg. Technol.* 86, 580–596. <https://doi.org/10.1016/J.TRC.2017.11.027>.
- Zhang, Z., Yang, W., Wushour, S., 2020. Traffic accident prediction based on LSTM-GBRT model. *J. Control Sci. Eng.* 1–10. <https://doi.org/10.1155/2020/4206919>.
- Zhao, H., Mao, T., Duan, J., Wang, Y., Zhu, H., 2019. FMCNN: a factorization machine combined neural network for driving safety prediction in vehicular communication. *IEEE Access* 7, 11698–11706. <https://doi.org/10.1109/ACCESS.2019.2891619>.
- Zhao, H., Zhang, J., Li, X., Wang, Q., Zhu, H., 2020. Deep learning-based prediction of traffic accident risk in vehicular networks. In: *2020 IEEE Globecom Workshops, GC Wkshps 2020 - Proceedings*. Institute of Electrical and Electronics Engineers Inc., pp. 11698–11706. <https://doi.org/10.1109/GCWKSHPS50303.2020.9367497>
- Zhao, H.Z., Rao, G., 2021. Traffic accident prediction methods based on multi-factor models. In: *International Conference on Knowledge Science, Engineering and Management (KSEM 2021)*. Springer Science and Business Media Deutschland GmbH, pp. 41–52. https://doi.org/10.1007/978-3-030-82153-1_4.
- Zheng, M., Li, T., Zhu, R., Chen, J., Ma, Z., Tang, M., Cui, Z., Wang, Z., 2019. Traffic accident's severity prediction: a deep-learning approach-based CNN network. *IEEE Access* 7, 39897–39910. <https://doi.org/10.1109/ACCESS.2019.2903319>.
- Zhu, S., Wang, K., Li, C., 2021. Crash injury severity prediction using an ordinal classification machine learning approach. *Int. J. Environ. Res. Publ. Health* 18, 11564. <https://doi.org/10.3390/IJERPH182111564>.
- Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., He, Q., 2021. A comprehensive survey on transfer learning. *Proc. IEEE* 109, 43–76. <https://doi.org/10.1109/JPROC.2020.3004555>.